

FIELD MAINTENANCE PRINT SET

TABLE OF CONTENTS	
LA8 FIELD MAINTENANCE PRINT SET	B-TC-LA8-P-5
DRAWING DIRECTORY	B-DD-LA8-P
LA180 INTERFACE PARTS LIST	A-PL-LA8-P-0
FIELD ACCEPTANCE AND INSTALLATION PROCEDURE	A-SP-LA8-P-1
SHIPPING LIST	A-PL-LA8-P-2
SOFTWARE LIST	A-PL-LA8-P-3
ENGINEERING SPECIFICATION	A-SP-LA8-P-4
LA180 INTERFACE	D-CS-M8365-0-1
INTERFACE CABLE (BC11S)	D-IA-BC11S-0-0
LA180 DRAWING DIRECTORY (COMPLETE SET)	B-DD-LA180-0

LA8-PA
LA8-PB
LA8-PC
LA8-PD
LA8-PK
LA8-PL
LA8-PM
LA8-PN

LA8

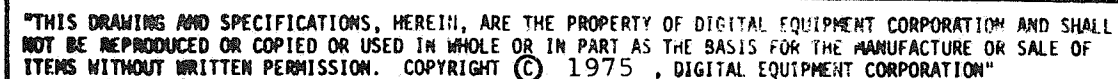
Field Maintenance

Print Set

Digital Equipment Corporation

PRINT SET ORDER NUMBER:
MP00075

[illegible]



THIS IS PRINT SET

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SEQUENCE 70

[illegible]

EN-01002-1A-16-N974-258

REVISIONS			
DATE	CHG. NO.	REV	

USED ON OPTION/MODEL		DRN.	DATE	TITLE			
		J. DEVIN	1-22-76				
		CHKD.	DATE				
		<i>[Signature]</i>	31-MAR-76				
		<i>[Signature]</i>	DATE	LA180 INTERFACE			
		<i>[Signature]</i>	31-MAR-76				
		PROD.	DATE	SIZE	CODE	NUMBER	REV
		2 Bornstein	7 APR 76	B	DD	LA8-P	
		FIELD CODE	DATE				
		<i>[Signature]</i>	31-MAR-76	DIST			
SHEET 1 OF 2							

CUSTOMER PRINT SET				ELECTRICAL					CUSTOMER PRINT SET				MECHANICAL								
				FIG. SET	FIND NO.	DRAWING NO.	REV	NO OF SHT	DESCRIPTION	OPTION NO./FILE DATE					FIG. SET	FIND NO.	DRAWING NO.	REV	NO OF SHT	DESCRIPTION	OPTION NO./FILE DATE
					1	A-PL-LA8-P-0		1	LA180 INTERFACE PARTS LIST						1	A-PL-LA8-P-0		1	LA180 INTERFACE PARTS LIST		
						A-SP-LA8-P-1			FIELD ACCEPT. AND INSTALL. PROC.												
						A-PL-LA8-P-2		1	SHIPPING LIST												
						A-PL-LA8-P-3		1	SOFTWARE LIST												
						A-SP-LA8-P-4		4	ENGINEERING SPECIFICATION												
						B-TC-LA8-P-5		1	FIELD MAINTENANCE PRINT SET												
						A-SP-LA8-P-6		5	LA8 IN-HOUSE ACCEPT. PROCED.												
					2	D-CS-M8365-0-1			LA180 INTERFACE FOR 8/E/F/M/A	M8365											
						K-CO-M8365-0-4			X-Y HOLE COORDINATE LOCATION												
						D-AH-M8365-0-5			ASSY/DRILLING HOLE LAYOUT												
						B-MH M8365-0-6			MODULE HISTORY												
					3	B-DD-LA180-0			LA180 DRAWING DIRECTORY	LA180					3	B-DD-LA180-0			LA180 DRAWING DIRECTORY	LA180	
					4	D-IA-BC11S-0-0		1	INTERFACE CABLE						4	D-IA-BC11S-0-0		1	INTERFACE CABLE		

CUSTOMER PRINT SET CODES

DRB 108

X = PRINT OF DOCUMENT INCLUDED IN FIRST SET

C = INCLUDES ALL PRINTS INDICATED ON DOCUMENT

S = CONFIDENTIAL AUTHORIZED SIGNATURE REQUIRED

TITLE: LA180 INTERFACE

SHEET 2 OF 2

SIZE CODE: B DD

NUMBER: LA8-P

REV:

DIGITAL EQUIPMENT CORPORATION
MAYNARD, MASSACHUSETTS

ENGINEERING SPECIFICATION

DATE 3/31/76

TITLE FIELD INSTALLATION & ACCEPTANCE PROCEDURE FOR LA8

REV

DESCRIPTION

CHG NO

ORIG

DATE

APPD BY

DATE

ENG Carl Cline

APPD 11-1-76

CHG NO 31-NOV-76

SIZE CODE A SP

NUMBER LA8-P-1

REV

DEC FORM NO EN-01022-16N376-1A31

JRA 100

1 of 3

ENGINEERING SPECIFICATION

CONTINUATION SHEET

TITLE FIELD INSTALLATION & ACCEPTANCE PROCEDURE FOR LA8

I. GENERAL

This procedure defines the performance standards required of the following LA8 configurations:

LA8-PA	LA180 and M8365	115V	60 Hz	25 ft. cable
LA8-PB	LA180 and M8365	230V	60 Hz	25 ft. cable
LA8-PC	LA180 and M8365	115V	50 Hz	25 ft. cable
LA8-PD	LA180 and M8365	230V	50 Hz	25 ft. cable
LA8-PK	LA180 and M8365	115V	60 Hz	100 ft. cable
LA8-PL	LA180 and M8365	230V	60 Hz	100 ft. cable
LA8-PM	LA180 and M8365	115V	50 Hz	100 ft. cable
LA8-PN	LA180 and M8365	230V	50 Hz	100 ft. cable

II. INSPECTION

After removing the M8365 from the packing material, inspect the LA8 for the following:

- Loose or broken components
- Inventory the hardware against the shipping list
- Inventory the prints against the shipping list, if ordered
- Inventory the software against the software list, if ordered

III. INSTALLATION PROCEDURE

Before installing the M8365 into the OMNIBUS, insure that the desired device code is selected (65 or 66).

Device Code 66	=	Jumper 1	Jumper 2
Device Code 65	=	A to B	C to D
		A to D	B to C

Installing the M8365 into a PDP8A:

- Remove the consoles from the front of the 8A.
- Route the BC11-S cable end, marked "Pl", through the access hole at the rear of the machine and connect it to the M8365.

DEC FORM NO EN-01022-16N376-1A31

JRA 100

SHEET 2 OF 3

ENGINEERING SPECIFICATION

CONTINUATION SHEET

TITLE FIELD INSTALLATION & ACCEPTANCE PROCEDURE FOR LA8

3. Insert the M8365 in the lowest available vacant slot in the OMNIBUS.

4. Dress the BC11-S properly in the 8A and replace the consoles.

Installing the M8365 into a PDP8E/M/P:

1. Remove the top cover.

2. Route the BC11-S cable end, marked "Pl", through the rear of the machine and connect it to the M8365.

3. Insert the M8365 into any vacant slot in the OMNIBUS.

4. Dress the BC11-S and strain relief the cable at the rear of the machine.

5. Replace the top cover.

Before connecting the BC11-S cable to the LA180, insure that the printer can run in the self test mode. Refer to LA180 User's Manual.

IV. ACCEPTANCE PROCEDURE

Perform the following procedure using the MAINDEC-08-DILAC.

- Run one pass of the Operator's Intervention Tests.
Starting address 200, SR=0000. Approximate time 10 min.
- Run several passes of tests 20 through 31. Starting address 001, SR=0000. Approximate time 15 min.
- Run test 60 for approximately 15 minutes. Approximate time 15 min.

Refer to the MAINDEC listing for additional information regarding error reporting and operator usage.

Other reference material that may be required:

- LA180 User's Manual EK-LA180-Op
- LA8 Printset MP-00075
- LA8 Maintenance Manual EK-LA008-MM

DEC FORM NO EN-01022-16N376-1A31

JRA 100

SHEET 3 OF 3

[illegible]

DIGITAL EQUIPMENT CORPORATION
MAYNARD, MASSACHUSETTS

ENGINEERING SPECIFICATION

TITLELAB-XX

DATE12/22/75

REV

DESCRIPTION

CHG NO

ORIG

DATE

APPD BY

DATE

ENG John K. Blawie

APPD John K. Blawie

SIZEA

CODESP

NUMBERLAB-P-4

REV

DEC 16 (592)-1079A-R873

DRA 107A

SHEET1OF4

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ENGINEERING SPECIFICATION

TITLELAB-XX

GENERAL:

The LAB-XX is an OMNIBUS compatible controller incorporating the LA180 DECprinter in its various voltage and frequency variations. The controller is a single quad card which plugs into the OMNIBUS of a PDP-8/M/A.

VARIATIONS:

LAB-PA - M8365, BC11S-25, LA180PA 117V 60 HZ
LAB-PB - M8365, BC11S-25, LA180PB 230V 60 HZ
LAB-PC - M8365, BC11S-25, LA180PC 117V 50 HZ
LAB-PD - M8365, BC11S-25, LA180PD 230V 50 HZ
LAB-PK - M8365, BC11S-A0, LA180PA 117V 60 HZ
LAB-PL - M8365, BC11S-A0, LA180PB 230V 60 HZ
LAB-PN - M8365, BC11S-A0, LA180PC 117V 50 HZ
LAB-PD 230V 50 HZ

PHYSICAL:

A single quad card (M8365), a 25 or 100 foot cable and the appropriate LA180 variation.

ELECTRICAL:

+5 volts @ 0.65 amps (controller only, the LA180 requires a separate source of 117/230 volts A.C.).

PROGRAMMING:

The LAB is programmed as the LS8 E/F or KL8 series serial line controllers (output only). The interpretation of data may, however, be different between the LA180 and other printing devices.

6660 - Set the print flag, an internal instruction requiring no LA180 interaction, directly sets the print flag.

SIZEA

CODESP

NUMBERLAB-P-4

REV

DEC FORM NO DEC 16-(381)-1022-N370

DRA 108

SHEET2OF4

ENGINEERING SPECIFICATION

TITLELAB-XX

6661 - Skip on print flag, if the print flag is set, the next sequential instruction is skipped.

6662 - Clear the printer flag, at TP3 of this instruction the print flag is directly cleared.

6663 - Unused

6664 - Load the printer buffer, AC 5-11 is transferred to a buffer internal to the M8365 module and a transfer flag is set. If the LA180 is online, a strobe pulse is issued, starting at TP4 of this instruction and of length 250 nanosecond (+/-20%). If the LA180 is not online, this transfer flag remains set. Setting the LA180 online will initiate the transfer.

6665 - Set or clear the interface interrupt enable. If AC11 (1) when this instruction is issued interrupt enable is set, if AC11 (0), interrupt is cleared.

6666 - A combination of 6662 and 6664.

6667 - Unused

INITIALIZATION:

Powering up the computer into which the M8365 is plugged, depressing the INIT button (initialize key), or issuing a CAF instruction causes the following:
Interrupt Enable is set.
The print flag is cleared.
The Internal Data Buffer is cleared.

INTERCONNECTION:

The LAB-XX has either a 25 foot or 100 foot cable to the LA180. Applications requiring cable lengths greater than 100 feet should be referred to Digital's Computer Special Systems Group.
The connector on the M8365 module is a 40 pin Berg type, see Table 1 for pinout details.

SIZEA

CODESP

NUMBERLAB-P-4

REV

DEC FORM NO DEC 16-(381)-1022-N370

DRA 108

SHEET3OF4

ENGINEERING SPECIFICATION

TITLELAB-XX

Device Code

66 - Standard

65 - Optional, jumper selectable

Environment

See PDP 8/A specification (Dec Std. 102-C)

Table 1

Signal Name pin

Demand X

Gnd W

Strobe VV

Gnd UU

Online N

Gnd M

Data MSB (P07) DD

Gnd CC

Data (P06) RR

Gnd PP

Data (P05) TT

Gnd SS

Data (P04) FF

Gnd EE

Data (P03) BB

Gnd AA

Data (P02) LL

Gnd KK

Data LSB (P01) JJ

Gnd HH

See DEC part #12-09941-02 for pin definitions.

SIZEA

CODESP

NUMBERLAB-P-4

REV

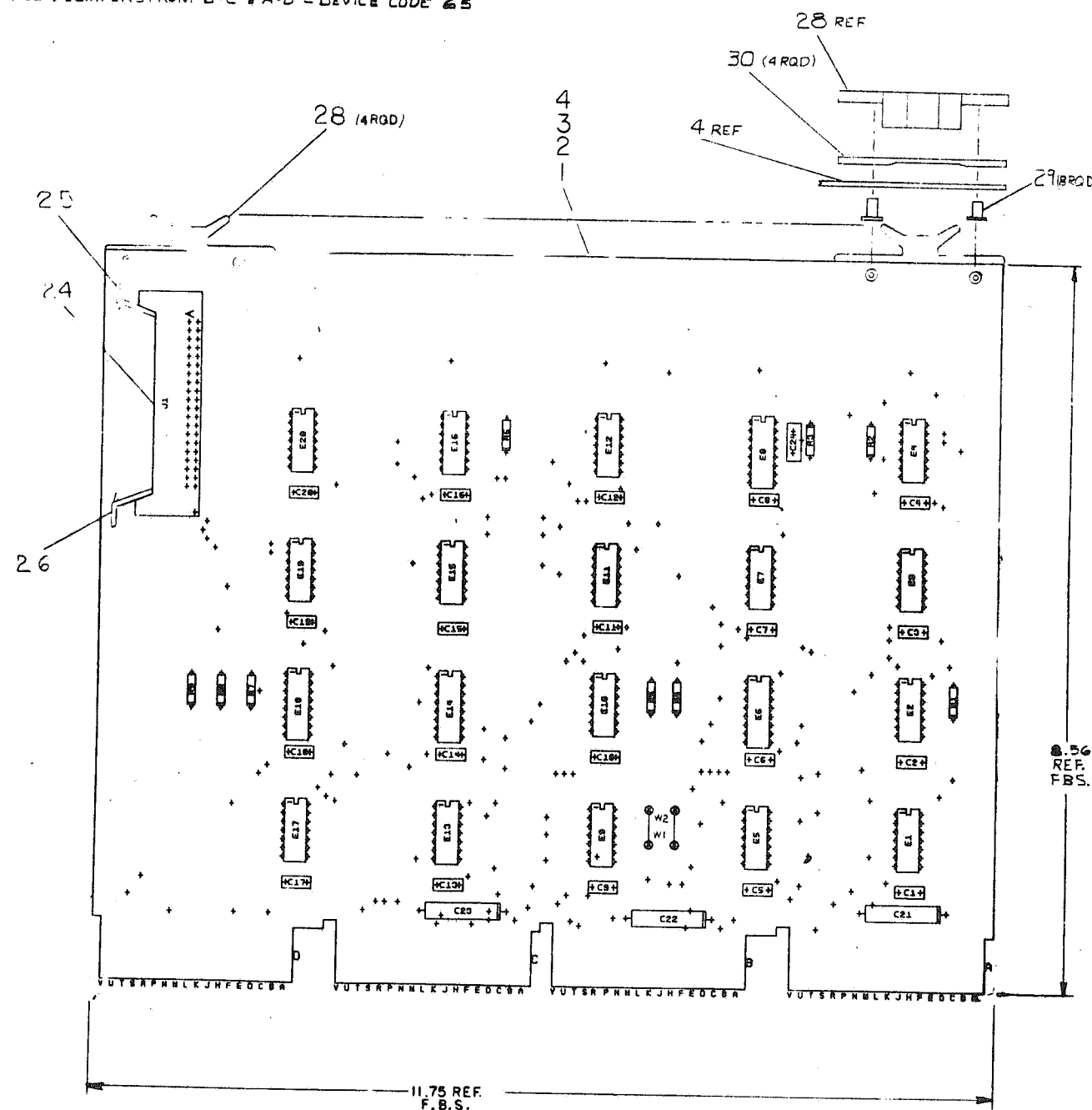
DEC FORM NO DEC 16-(381)-1022-N370

DRA 108

SHEET4OF4

1 STANDARD JUMPER CONFIGURATION IS W1 IN, W2 IN - DEVICE CODE 66
OPTION IS W1, W2 OUT, JUMPERS FROM B-C & A-D - DEVICE CODE 65

PART CALLED FOR			SUBSTITUTE PART		
QTY	PART NO.	DESCR.	QTY	PART NO.	DESCR.
20	1001610-01	0.0146 DISC	20	1001610-00	0.0146 GLASS
1	1909708	314	1	1910391	5314
1	709486	384	1	1910399	5384
1	1909486	384	1	1910393	7384
4	711169	8640	4	1910392	5380
4	1909771	6380	4	1910390	7380
4	1911467	8640	4	191115	11380
3	1909705	8881	3	1909773	97401



AA2, BA2, CA2, +5V

C21, 22, 23 6,8 μ F

C1-C20 0,1 μ F

AC1, AC2, AF1, AF2, AN1, AN2
AT1, AT2, BC1, BC2, BF1, BF2
BN1, BN2, DT1, DT2, CC1, CC2
CF1, CF2, CN1, CN2, CT1, CT2,
DC1, DC2, DF1, DF2, DN1, DN2
DT1, DT2.

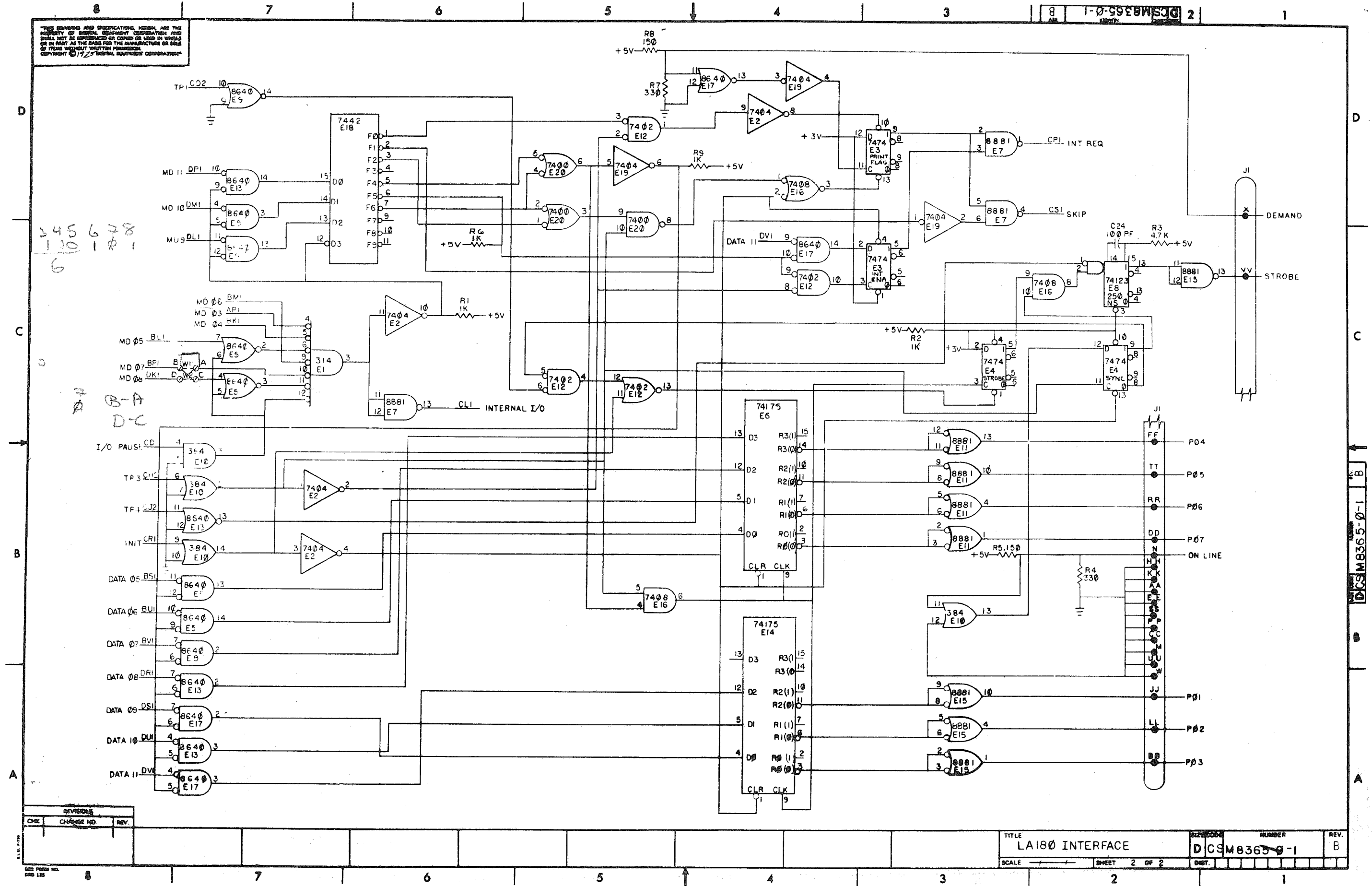
[illegible]

REF.	DESCRIPTION	PART NUMBER	QTY
REF.	X Y COORDINATE HOLE LOCATION	K-CD-M8365-Ø-4	1
REF.	ASSY/DRILLING HOLE LAYOUT	D-AH-M8365-Ø-5	2
REF.	MODULE ECO HISTORY	B-MH-M8365-Ø-6	3
1	ETCHED CIRCUIT BOARD	5011818	4
1 C24	CAP, 100PF 100V, 5%	1000016	5
20 C1 THRU C20	CAP, 0.1µF, 30V, 20%	1001610-01	6
3 C21, C22, C23	CAP, 6.8µF, 35V, 10%	1005306	7
2 R5, R8	RES 150 1/4W 5%	1300250	8
2 R4, R7	RES 330 1/4W 5%	1300295	9
4 R1, R2, R6, R9	RES 1K 1/4W 5%	1300365	10
1 R3	RES 4.7K, 1/4W 5%	1300447	11
2 E3 E4	DEC IC 7474	1905547	12
1 E20	DEC IC 74130	1905575	13
1 E12	DEC IC 7402	1909004	14
1 E10	DEC IC 364	1909486	15
2 E2 E19	DEC IC 7404	1909486	16
1 E1	DEC IC 314	1909704	17
3 E7, E11 E15	DEC IC 6881	1909705	18
1 E18	DEC IC 7442	1910046	19
1 E16	DEC IC 7404	1910155	20
1 E8	DEC IC 74123	1910436	21
2 E6 E14	DEC IC 74175	1910651	22
4 E5, E9, E13, E17	DEC IC 8640	1911469	23
1 J1	CONN 40 PIN RT ANGLE HEADER	1209941-02	24
1	LATCH L H	1209941-03	25
1	LATCH R H	1209941-04	26
4	SPLIT LUGS	9006735	27
4	HANDLE FLIP CHIP	9008337-06	28
8	EYELET	9006732	29
4	SPACER (CABLE CLAMP)	1202704	30
A/W	W1, W2	WIRE, 22 AWG, SOLID	31

QTY	REF. DESIGNATION										DESCRIPTION										PART NO.										REV.																		
PARTS LIST																																																	
ETCH BOARD REV.										B																																							
										DRAWN BY <i>[Signature]</i>										DATE 3/7/68										00-0000 TITLE LA180 INTERFACE																			
										CHECKED BY <i>[Signature]</i>										SHEET 1 OF 15																													
										ENG. <i>[Signature]</i>										DATE 18 Nov 75																													
										MFGT. ENG. <i>[Signature]</i>										DATE 18 Nov 75																													
										APP. <i>[Signature]</i>										DATE 18 Nov 75																													
										NEXT MEMBER ASSY																																							
DEC NO.										EIA NO.										B-DD-PDP-8/E										SCALE <i>[Signature]</i>										REV. B									
CONVERSION CHART																																																	
INCHES 1 OF 2										CS M8365-0 - I										REV. B																													

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1-0-090895-00 2



REVISIONS		
CHK	CHANGE NO.	REV.

TITLE		SIZE		NUMBER		REV.	
LA180 INTERFACE		D		CSM8365-0-1		B	
SCALE		SHEET 2 OF 2		DWT.			

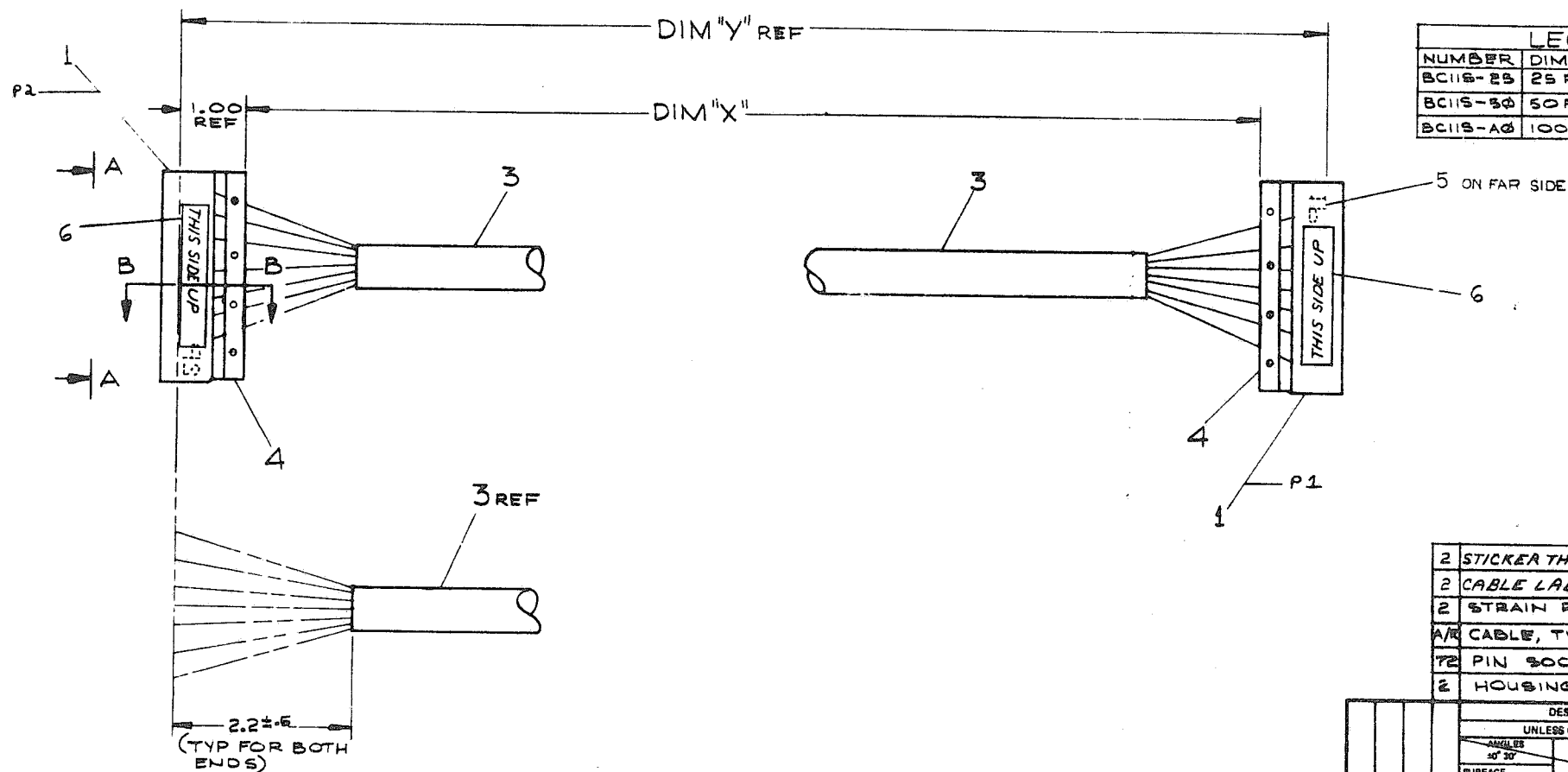
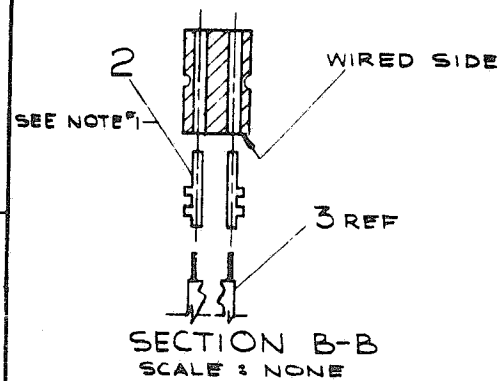
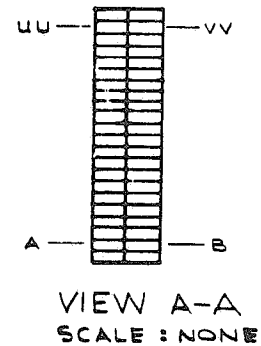
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WIRE TABLE															
FROM				TO				FROM				TO			
ITEM NO	AWG	COLOR	CONNECTION	WITH	CONNECTION	WITH	SIGNAL NAME	ITEM NO	AWG	COLOR	CONNECTION	WITH	CONNECTION	WITH	SIGNAL NAME
3	24	BLU	PI-SS	2	P2-D	2	D5 RTN	3	24	GRN	PI-V	2	P2-AA	2	PAPER GND
	TWP	WHT	PI-PP		P2-F		D6 RTN		TWP	RED	PI-U		P2-BB		PAPER RTN
		GRN	PI-AA		P2-V		D3 RTN			BLU	PI-VV		P2-A		STROBE
		WHT	PI-EE		P2-R		D4 RTN			BLK	PI-UU		P2-B		STROBE RTN
		GRN	PI-L		P2-KK		* +S			ORN	PI-N		P2-HH		SELECT (ONLINE)
		WHT	PI-K		P2-LL		* RTN			BLK	PI-M		P2-JJ		SELECT RTN
		BRN	PI-HH		P2-N		D RTN			GRN	PI-T		P2-CC		HDWR
		WHT	PI-KK		P2-L		D2 RTN			BLK	PI-S		P2-DD		HDWR RTN
		GRY	PI-F		P2-PP		*SEE NOTE 2			BRN	PI-Z		P2-W		* SEE NOTE 2
		WHT	PI-E		P2-RR		*SEE NOTE 2			BLK	PI-Y		P2-X		* SEE NOTE 2
		BLU	PI-NN		P2-H		PRIME			GRY	PI-DD		P2-S		D7
		RED	PI-MM		P2-J		PRIME RTN			BLK	PI-D		P2-SS		DB(PARITY)
		GRN	PI-CC		P2-T		D7 RTN			BLU	PI-TT		P2-C		D5
		RED	PI-C		P2-TT		D8 RTN			YEL	PI-RP		P2-E		D6
		GRN	PI-R		P2-EE		FAULT			ORN	PI-BB		P2-U		D3
		RED	PI-P		P2-FF		FAULT RTN			YEL	PI-FF		P2-P		D4
		BRN	PI-X		P2-Y		DEM			GRN	PI-JJ		P2-M		D1
	24		PI-W	2	P2-Z	2	DEM RTN	3	TWP	YEL	PI-LL	2	P2-K	2	D2

NOTES:

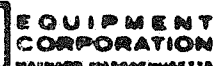
1. MFG. SHOULD USE MACHINE CRIMPING TOOL FOR THE ASSEMBLY OF PINS (ITEM # 2)
2. UNUSED CROWN

LEGEND		
NUMBER	DIM "Y" VARIATION	DIM "Y" (PRECUT) REF
BC115-25	25 FT. ± 3.0 IN	25 FT. 2.0 IN
BC115-50	50 FT. ± 2.0 IN	50 FT. 2.0 IN
BC115-A0	100 FT. ± 2.0 IN	100 FT. 2.0 IN



DESCRIPTION	DWG./PART NO.	ITEM NO.
2 STICKER THIS SIDE UP	3611567	6
2 CABLE LABEL	9008473	5
2 STRAIN RELIEF	1211166	4
1/4 CABLE, TWP #24 AWG	9107700-00	3
72 PIN SOCKET	1210089-4	2
2 HOUSING	1210918-15	1

THIRD ANGLE PROJECTION	DATE 1/10/75	FIRST USED ON digital
CHK'D 1/10/75	ENG 1/10/75	PROJ. ENG 1/10/75
DO NOT SCALE DWG	NEXT HIGHER ASSY.	
MATERIAL	A-PL-CMIF-0-0	SIZE CODE
FINISH	SCALE	NUMBER
	SHEET 1 OF 1	DIST.



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THIS IS PRINT SET []

RECEIVED

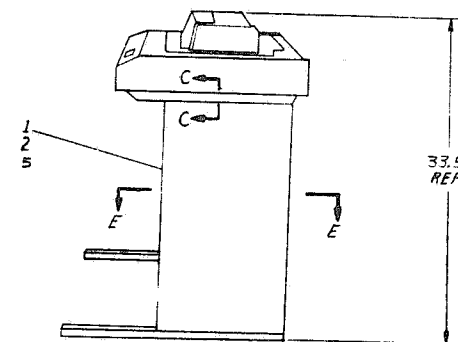
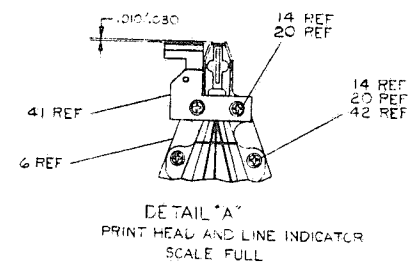
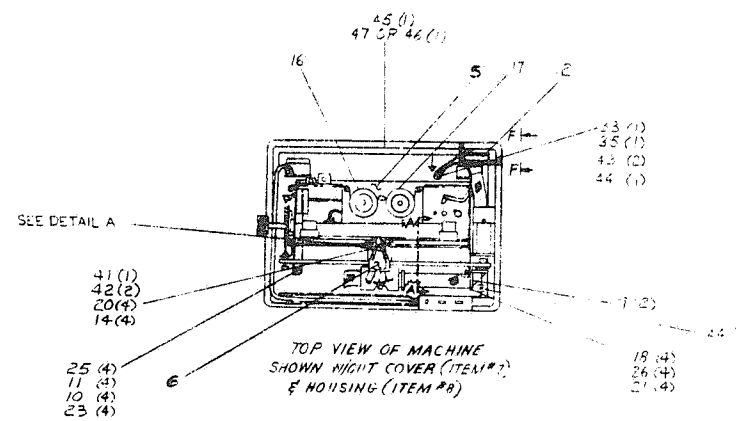
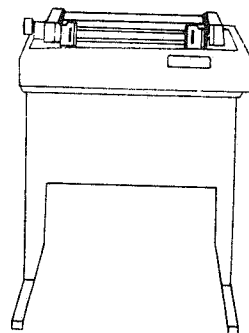
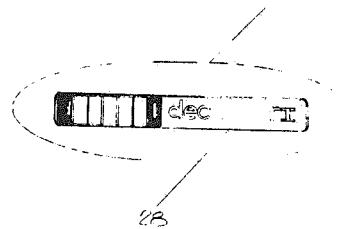
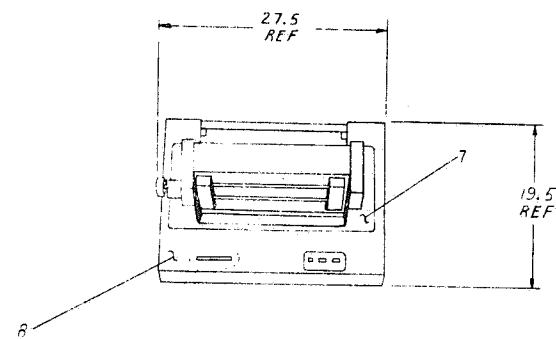
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D-CS-5411653-0-1
D-IC-7011167-0-0
A-SP-3700131-0-0
B-DD-LAXX-KB
B-DD-LAXX-LD
B-DD-LAXX-KJ

[illegible]

REV		REV	
2-78	LA 180-22	B	C
2-78		LA 180-48	

USED ON OTHERS		NAME	DATE	TITLE DECRIPTER DRAWING DERECTORY
		A. TAYLOR	1/2/74	
		A. TAYLOR	6/18/75	
		E. STENTZ	6/17/75	
		E. STENTZ	6/18/75	
		E. STENTZ	6/18/75	
		T. MONELLO	6/23/75	

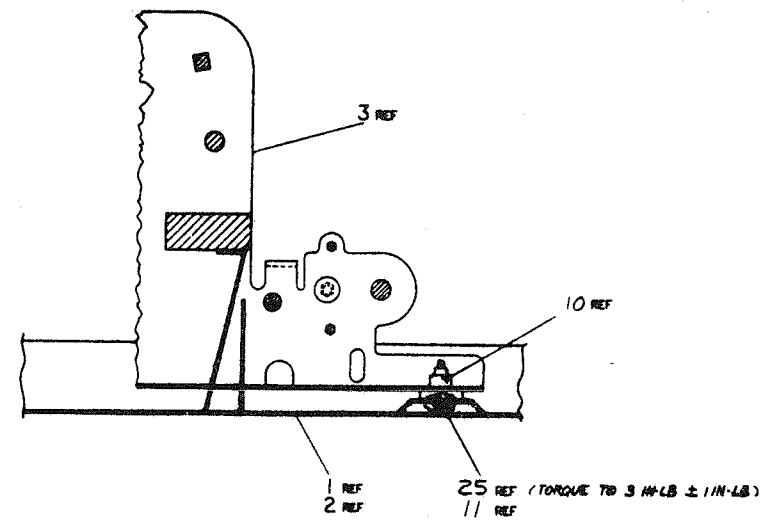
LA 180-6		C
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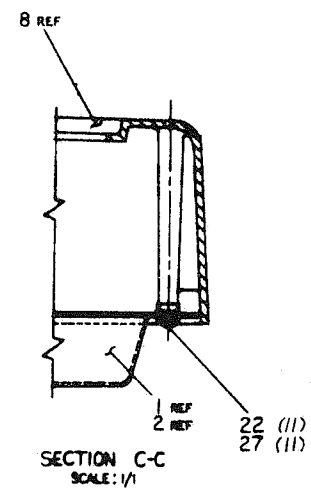
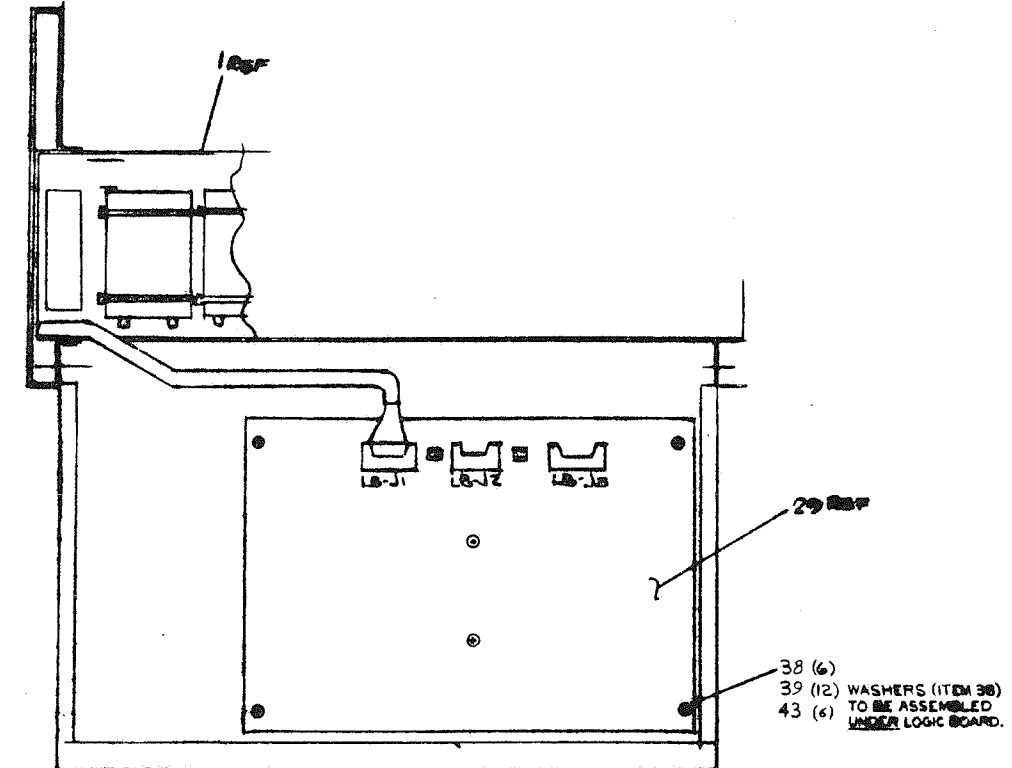
NOTES.

1. FOR PARTS LIST SEE DRAWING
"APL-LA 180-00"
2. ITEMS "32 (PLATE, INPUT CLAMP), "33 (2
SCREWS), "34 (2 LOCKWASHERS), "35 (2 PLAY-
WASHERS) AND "36 (2 TRAIN RELIEF) ARE TO BE
SHIPPED LOOSE WITH JN 7.

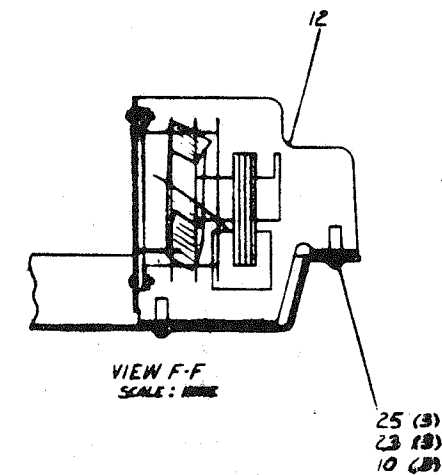
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SECTION A-A
SCALE: 1/1



SECTION C-C
SCALE: 1/1



VIEW F-F
SCALE: 1/1

WIRE TABLE							
ITEM NO	DESCRIPTION	FROM		TO		REMARKS	
		AWG	COLOR	CONN	TERM		CONN
5				MTR2-P2		PB-J5	
62				CH-P7		PC-J7	
62				CH-S6-1		S6-1	TOP
62				CH-S6-2		S6-3	SET
5				MTR1-P3		CH-A6	
5				PH-P1		PB-J6	
5				MTR1-P1		PB-J4	
5				CH-S7-1		CH-S7-3	
5				CH-S7-2		CH-S7-3	
5				CH-S7-3		CH-S7-1	
5				CH-S7-4		CH-S7-5	
5				CH-S7-5		CH-S7-4	
5				CH-S7-6		CH-S7-7	
5				CH-S7-7		CH-S7-6	
5				CH-S7-8		CH-S7-9	
5				CH-S7-9		CH-S7-8	
5				CH-S7-10		CH-S7-11	
5				CH-S7-11		CH-S7-10	
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DIGITAL EQUIPMENT CORPORATION MAYNARD, MASSACHUSETTS					QUANTITY / VARIATION													
PARTS LIST					1A180-PA	1A180-PB	1A180-PC	1A180-PD										
MADE BY J. TAYLOR		CHECKED		SECTION														
DATE 10/4/74		DATE 6/17/75		1														
ENG 6/17/75		PROD R.L. Jodanis		ISSUED SECT.														
DATE 6-17-75		DATE 6/18/75		1														
ITEM NO	DWG NO. / PART NO.		DESCRIPTION															
1	E-AD-7010094-1-0	CABINET ASSY W/POWER SUPPLY(115V) 60HZ		1	-	-	-											
2	E-AD-7010094-2-0	CABINET ASSY W/POWER SUPPLY(230V) 60HZ		-	1	-	-											
3	E-AD-7010094-3-0	CABINET ASSY W/POWER SUPPLY(115V) 60HZ		-	-	1	-											
4	E-AD-7010094-4-0	CABINET ASSY W/POWER SUPPLY(230V) 50HZ		-	-	-	1											
5	E-AD-7010091-0-0	PRINTER MECHANISM ASSY		1	1	1	1											
6	D-AD-7009882-0-0	PRINT HEAD ASSY		1	1	1	1											
7	D-IA-7010758-0-0	COVER ASSY		1	1	1	1											
8	E-IA-7010757-0-0	HOUSING ASSY		1	1	1	1											
9	1202704-00	CLAMP, CABLE		2	2	2	2											
10	9008896-00	WELNUT #10-32		7	7	7	7											
11	9007906-00	WASHER, SPLIT LOCK, #10		4	4	4	4											
12	C-AD-7010756-0-0	FAN ASSY 115V, MOTOR		1	1	1	1											
13	9007649-00	WASHER, EXT TOOTH LOCK, #6		1	1	1	1											
14	9006633-00	WASHER, INTERNAL TOOTH LOCK #6		4	4	4	4											
15	A-MD-7412103-0-0	PKG. OF PAPER (100 SHEETS)		1	1	1	1											
16	3612153-00	RIBBON SPOOL W/RIBBON		1	1	1	1											
17	1212008-00	RIBBON SPOOL (EMPTY)		1	1	1	1											
18	9006011-1	SCREW, PHL, PAN HD, #4-40 x .38L		4	4	4	4											
19	9006077-1	SCREW, PHL, PAN HD, #6-32 x .38L		1	1	1	1											
20	9007793-1	SCREW, PHL, PAN HD, #6-32 x .56L		4	4	4	4											
21	9006557-00	NUT, KEPS, #4-40		4	4	4	4											
22	9009630-01	SCREW, PHL, PAN HD, #6-18 x .50, SELF TH		11	11	11	11											

TITLE

1A180 UNIT ASSY

ASSY NO.

E-UA-1A180-0-0

SHEET 1 OF 3

SIZE CODE

A PL

DIST

NUMBER

1A180-0-0

REV.

F

ECO NO

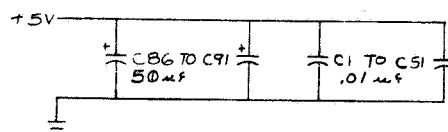
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DIGITAL EQUIPMENT CORPORATION MAYNARD, MASSACHUSETTS					QUANTITY / VARIATION															
PARTS LIST					LA180-PA	LA180-PB	LA180-PC	LA180-PD												
MADE BY DICK VENO		CHECKED		SECTION																
DATE 3/25/74		DATE 5-17-75		1																
ENG		PROD R. J. Godwin		ISSUED SECT.																
DATE 6-17-75		DATE 6/1/75		1																
ITEM NO.	DWG NO. / PART NO.	DESCRIPTION			LA180-PA	LA180-PB	LA180-PC	LA180-PD												
23	9006664-00	WASHER, FLAT, #10			7	7	7	7												
24	9007801-00	WASHER, SPLIT LOCK, #8			1	1	1	1												
25	9008955-1	SCREW, PHL PAN HD, #10-32 x .961			7	7	7	7												
26	9006658-00	WASHER, FLAT, #4			4	4	4	4												
27	9006653-00	WASHER, FLAT, #6			15	15	15	15												
28	C-PS-3612451-0-0	NAME PLATE			1	1	1	1												
29	D-CS-5411023-0-1	LOGIC BOARD LA180			1	1	1	1												
30	D-CS-5411023-PA-1	LOGIC BOARD LA180 C/C			-	-	-	-												
31	D-IA-7013469-0-0	C/C INTERFACE CABLE ASSY			-	-	-	-												
32	C-MD-7413216-0-0	PLATE, INPUT CLAMP			1	1	1	1												
33	9006037-1	SCR PHL PAN HD 8-32 x .38			3	3	3	3												
34	9006690-00	WASHER SPLIT LOCK #8			2	2	2	2												
35	9006660-00	WASHER FLAT #8			4	4	4	4												
36	9008509-00	STRAIN RELIEF			1	1	1	1												
37	A-SP-3700181-0-0	PACKAGING INSTRUCTIONS			REF	REF	REF	REF												
38	9008208-1	SCR PAN HD, NYLON 8-32 x .50			6	6	6	6												
39	9006708	WASHER NYLON .37 OD x .19 ID x .032 THK			12	12	12	12												
40	A-DC-7416178-0-0	DECAL, LA180			1	1	1	1												
41	C-MD-7414053-0-0	LINE INDICATOR			1	1	1	1												
42	9006656	WASHER, FLAT			2	2	2	2												
43	9006072-00	EXTERNAL TOOTH LOCKWASHER #8			8	8	8	8												
44	3612680-00	DECAL, GROUND SYMBOL			2	2	2	2												
TITLE LA180 UNIT ASSY				ASSY NO. E-UA-LA180-B-B	SIZE CODE A PL		NUMBER LA180-B-B		REV. ECO NO F											
SHEET 2 OF 2				DIST.																



89 TYP. 3 PLACES



GND AND 5V ARE USUALLY PIN 7 AND 14
RESPECTIVELY EXCEPTIONS ARE STATED ABOVE

IC PIN LOCATIONS

FIRST USED ON OPTION MODE

LA 180

SEMICONDUCTOR CONVERSION CHART

TITLE LA 180
LOGIC BOARD

DCS 5411023-0-

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PARTS LIST

QTY	REF DESIGNATION	DESCRIPTION	PART NO	ITEM NO
REF		K-Y COORDINATE HOLE LOCATION	K-CO-5411023-0-4	1
REF		ASSY/CELLING HOLE LAYOUT	K-AH-5411023-0-5	2
REF		MODULE ECO HISTORY	B-MH-5411023-0-6	3
1		ETCHED CIRCUIT BOARD	5011022-00	4
4	C79, C80, C81, C100	CAP 39pF 100V 5%	1000010-00	5
1	C85	CAP .68uF 50V 10%	1010031-06	6
2	C82, C83	CAP .047uF 100V 10%	1000051-00	7
6	C92, C94-97, C99	CAP .25uF 25V 10%	1000075-00	8
87	C1-C51, C71-E76	CAP .01uF 50V	1001610-00	9
1	C84	CAP .1uF 100V 10%	1002342-00	10
2	C77, C78	CAP .065uF 100V 20%	1001765-00	11
6	C86-C91	CAP .50uF 25V 10%	1001796-00	12
2	D1, D35	DIODE D664	1100114-00	13
2	D33, D34	DIODE 1N748A 3.9V 5%	1100122-00	14
2	D5, D6	DIODE 4001A 1.5V	1100123-00	15
1	D5	DIODE D662	1100115-00	16
3	J1, J2, J3	CONN 40 PIN	1205941-02	17
2	E1, E7	SOCKET IC 24 PIN	1210693-00	18
1	R86	RES 51K 1/4W 5%	1309422-00	19
14	R30-R39, R11, R24, R26, R27	RES 150 1/4W 5%	1300250-00	20
3	R78, R80, R82	RES 100 1/4W 5%	1300229-00	21
1	E57	RES 820 1/4W 5%	1301775-00	22
3	E59, E70, E71	RES 180 1/4W 5%	1301322-00	23
1	E83	RES 560 1/4W 5%	1301890-00	24
14	R40-R50, R33, R37, R10	RES 330 1/4W 5%	1300265-00	25
2	R84, R85	RES 470 1/4W 5%	1300316-00	26
24	E1-E11, E13-E17, R105, R11, R12, R59, E94, R27-R29, R106	RES 1K 1/4W 5%	1300365-00	27
1	R87	RES 1.5K 1/4W 5%	1300391-00	28
5	R81, R86, R90-E92	RES 51K 1/4W 5%	301574-50	29
2	R55, R56	RES 1K 1/4W 5%	1302748-00	30
1	E51	RES 47K 1/4W 5%	1302177-00	31
1	E51	RES 11.5K 1/4W 1%	1302060-00	32
2	E72, E73	RES 820K 1/4W 10%	1303187-00	33
2	R62, R63	RES 2.61K 1/4W 1%	1303303-00	34
2	E76, E77	RES 5.11K 1/4W 1%	1304854-00	35
				36
2	E74, E75	RES 5.9 1/4W 1%	1305726-00	37
1	R58	RES 316K 1/4W 1%	1305419-00	38
2	E52, E54	RES 225K 1/4W 1%	1305424-00	39
6	E64-E69	RES 2K 1/4W 5%	1302338-00	40
1	R60	RES 16.9K 1/4W 1%	1312618-00	41
1	E59	RES 24.3K 1/4W 1%	1309418-00	42
2	E2, E3	TRANS RS34D PNP	1503409-00	43
9	Q4-Q12	TRANS MPS-05 NPN	1510705-00	44
1	Q1	TRANS MPS-05 PNP	1510538-00	45
1	Y1	CRYSTAL 1.6896 MHE	1811689-00	46
11	E18, E20, E23, E25, E26, E27, E35, E50, E61, E54, E55	IC 7474	1905547-00	47
9	E24, E28, E29, E32	IC 7400	1905575-00	48
1	E29	IC 7410	1905576-00	49
1	E30	IC 7420	1905577-00	50
1	E31	IC 7401	1905590-00	51
2	E32, E33	IC 7493A	1909054-00	52

PARTS LIST

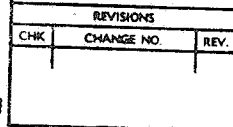
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3	E34, E41, E46	IC 8640	1911469-00	53
6	E38, E33, E47, E72, E75, E27	IC 7404	1909686-00	54
3	E45, E60, E73	IC 74154	1909701-00	55
1	E16	IC 7413	1909989-00	56
5	E12, E14, E62, E64, E67	IC 74193	1910018-00	57
1	E53	IC 7442	1910046-00	58
1	E19	IC 74190	1910095-00	59
1	E56	IC 8881	1909705-00	60
2	E57, E74	IC 74150	1910153-00	61
3	E5, E15, E21	IC LM301A	1910282-00	62
1	E65	IC 7489	1910396-00	63
1	E92	IC 74123	1910436-00	64
2	E34, E44	IC 74161	1910650-00	65
1	E22	IC 74175	1910651-00	66
2	E2, E8	IC 2101-2	2112323-00	67
2	E61, E66	IC 1702A PROMS	2300044-05	68
1	E38	IC 8815	1909713-00	69
4	E3, E9, E32, E37	IC 74197	1910035-00	70
4	E43, E57, E63, E68	IC 7408	1910155-00	71
1	E1	IC MK2626 (UNSLASHED O)	2307186-00	72
1	E7	IC MK2627 (ALPHA SET)	2307046-00	73
6		SPLIT LUGS	9006735	74
1		SCREW PPH	9006011-01	75
2		FASTON TAB	9007113-00	76
1		KEP NUT	9006557-00	77
2		EYLET	9009000	78
9	R98-R104, R109, R107	RES 47K 1/4W 5%	1302177	79
1	E58	IC 7414	1911324	80
1	E13	IC 74180	1910724	81
1	E10	IC 9601	1909373	82
1	E6	IC 7406	1910741	83
1	C98	CAP 39uF 10V 10%	1000076	84
7	R20 THRU R26	RES 4.7K 1/4W 5%	1300447-00	85
2	W1, W8	INSULATED JUMPER	9009185	86
AR		JUMPER, INSULATED	9105740-55	87
3		LATCH, LEFT	1209941-03	88
3		LATCH, RIGHT	1209941-04	89
1	E71	IC MK2611 IP CROM	23088A6	90

REVISIONS

DATE	CHANGE NO.	BY

TITLE	LA180 LOGIC BOARD	SIZE CODE	NUMBER	REV.
SCALE		SHEET 2 OF 9	DCS 5411023-0-1	R

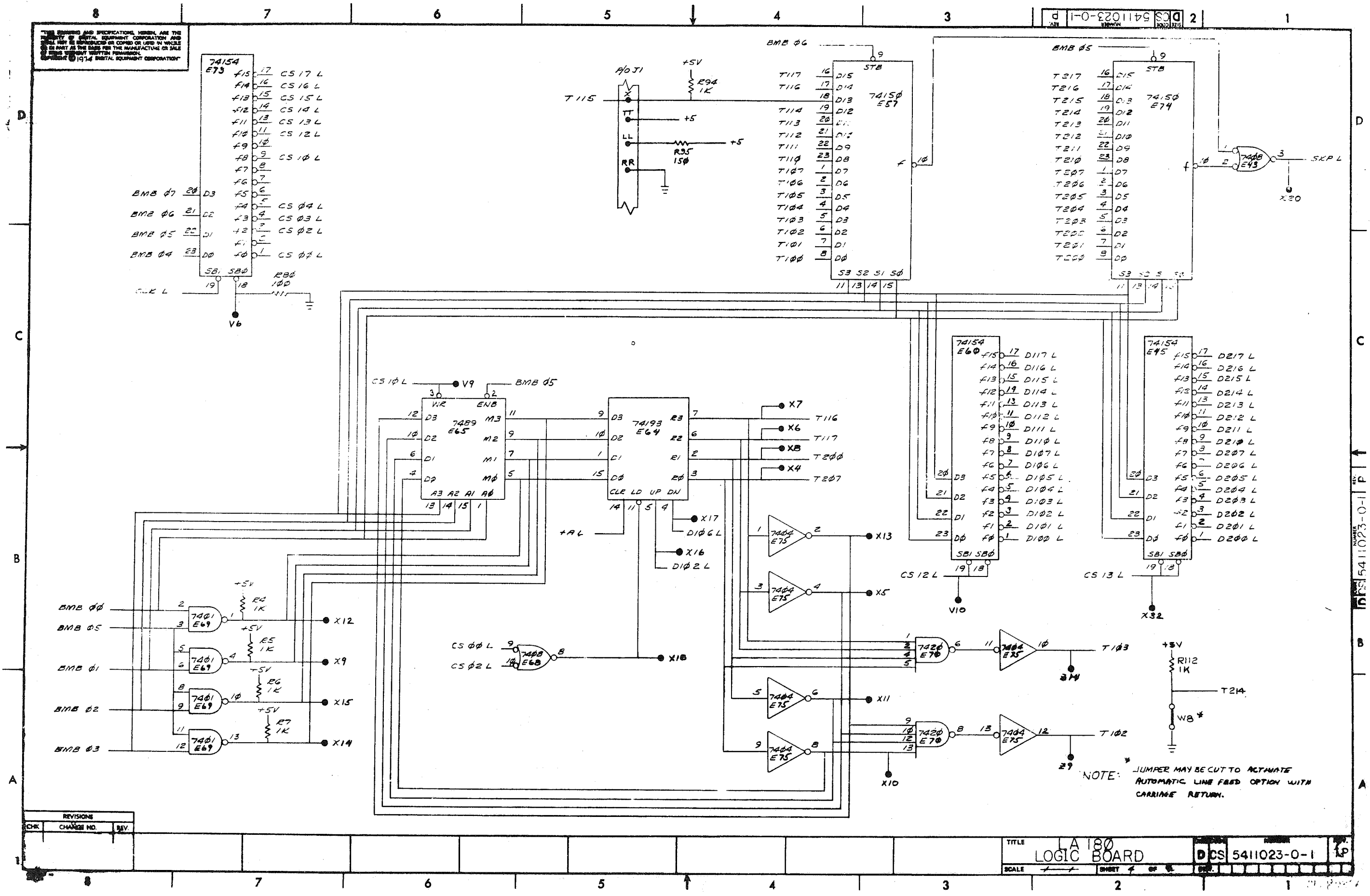
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SIZE CODE	NUMBER	REV
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UNIT		

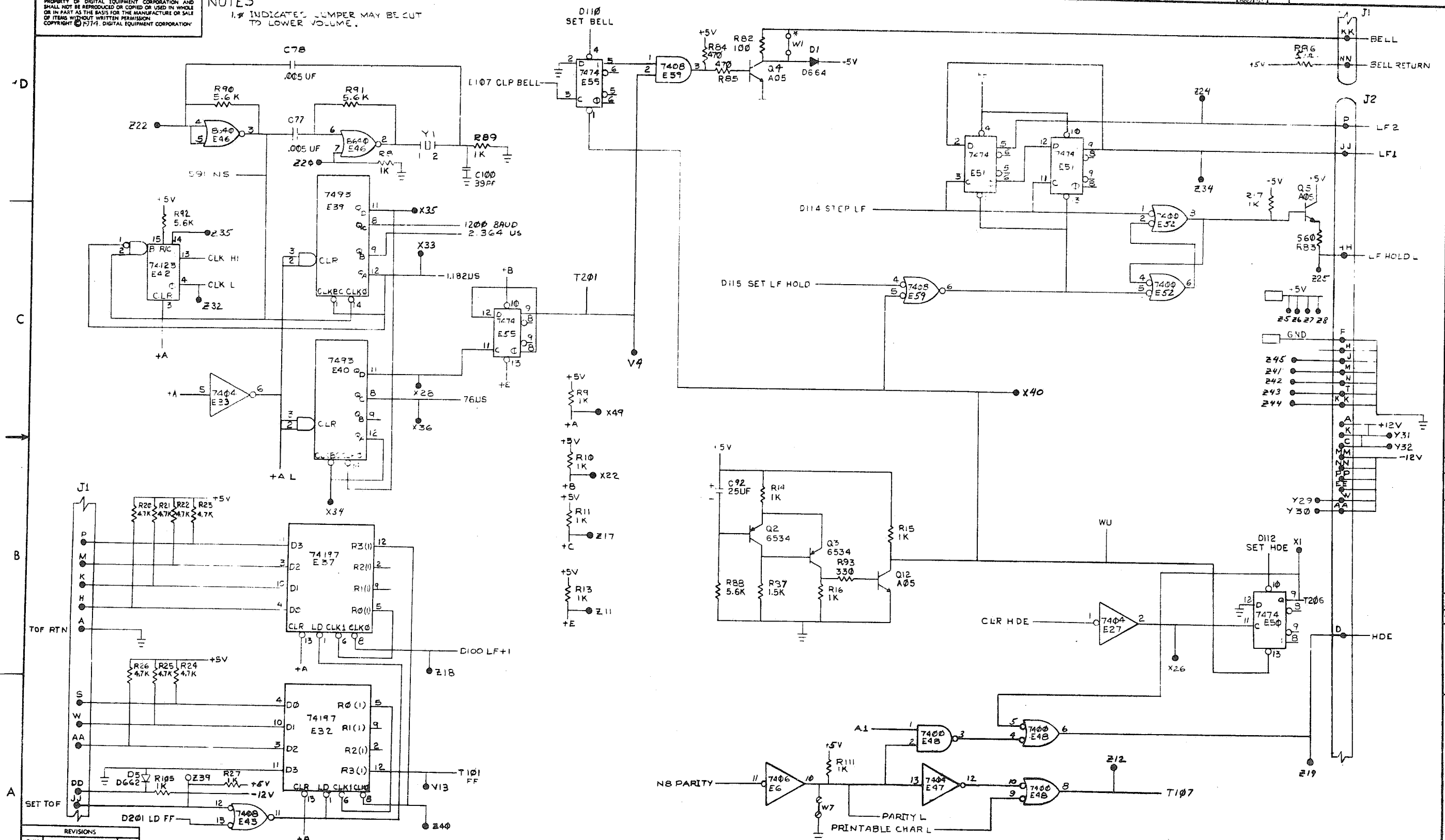
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DCS 5411023-0-1

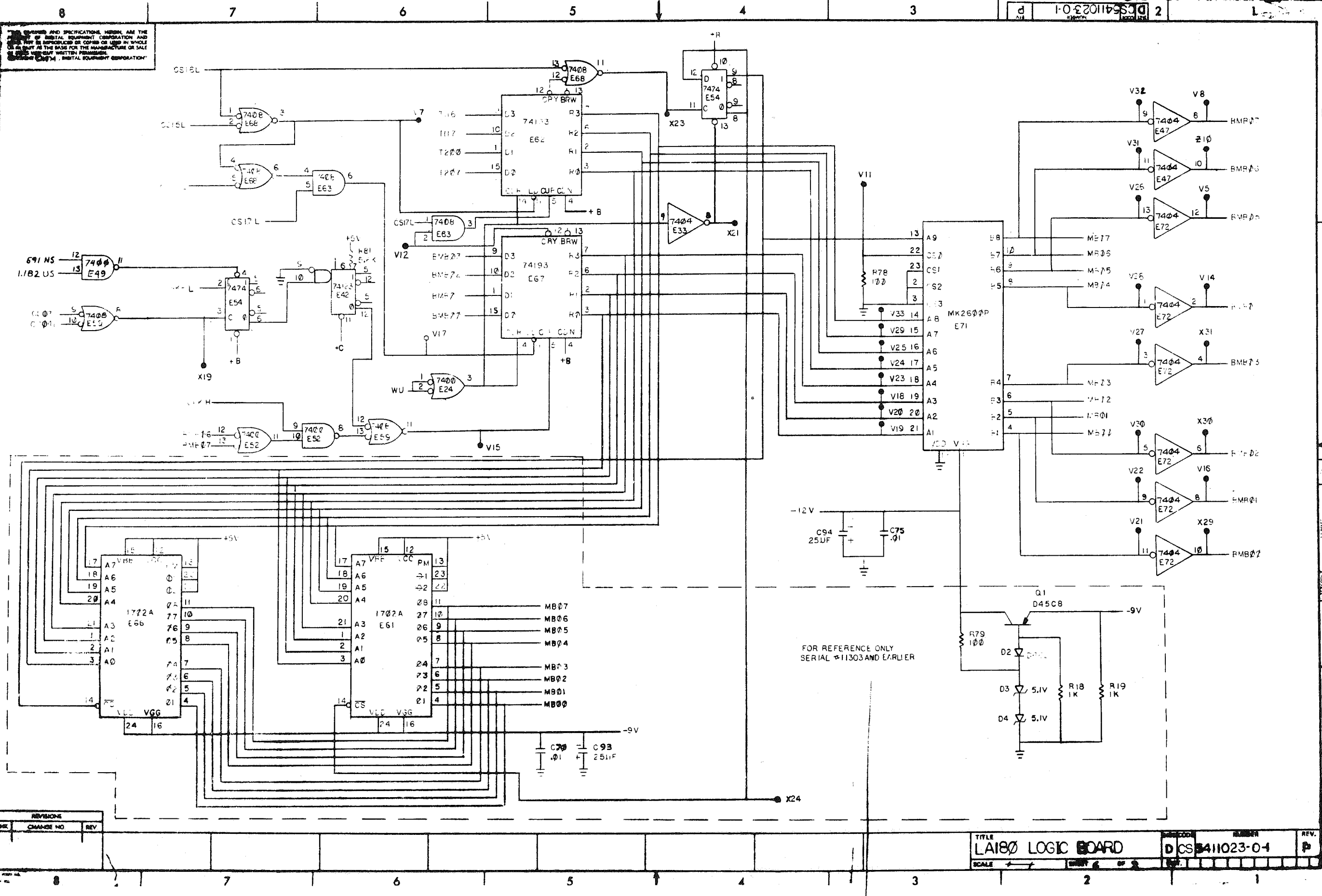


NOTE: JUMPER MAY BE CUT TO ACTIVATE AUTOMATIC LINE FEED OPTION WITH CARRIAGE RETURN.

REV.	CHG.	NO.	BY
1			

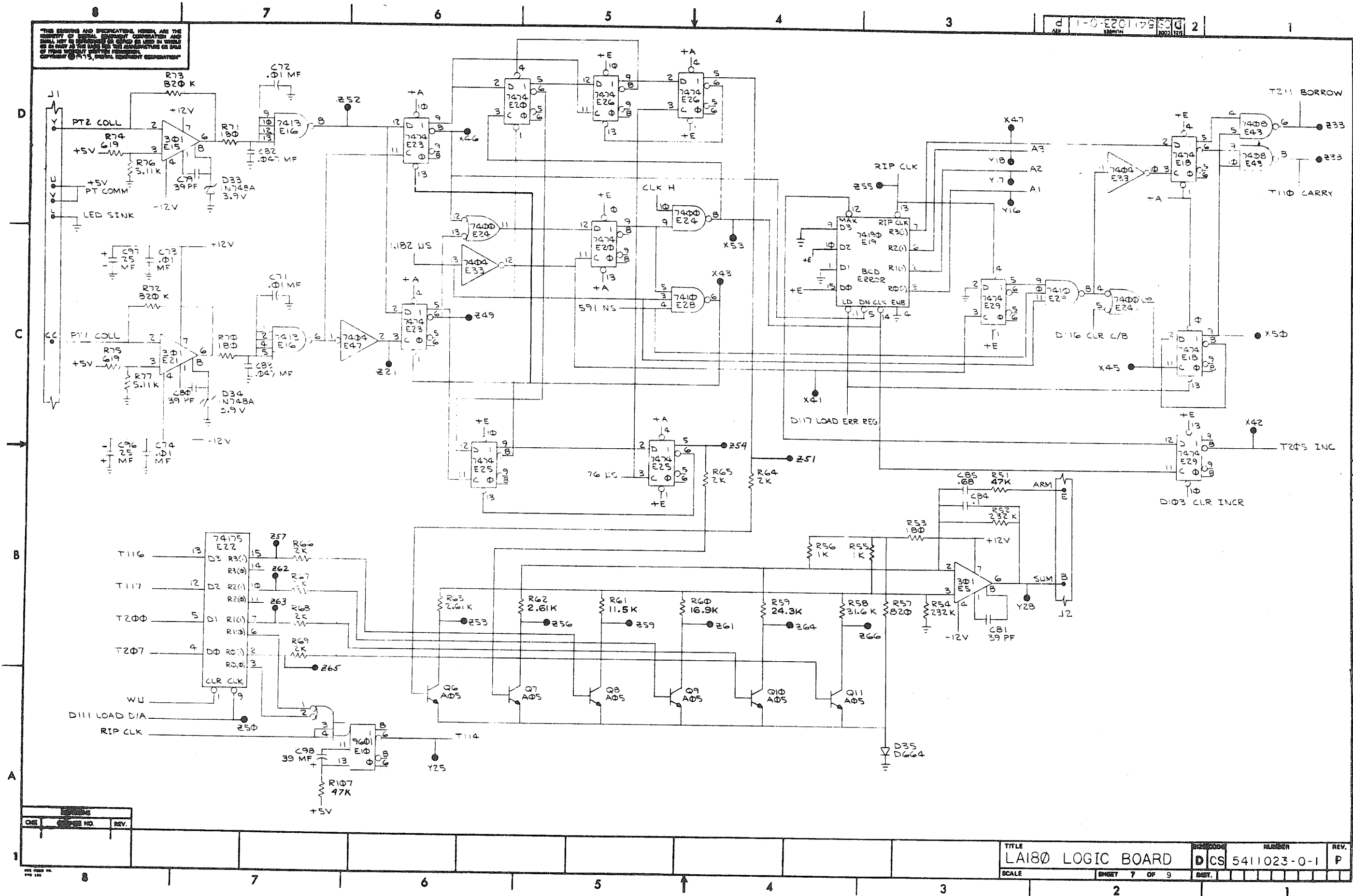


REVISIONS		
CHK	CHANGE NO.	REV.



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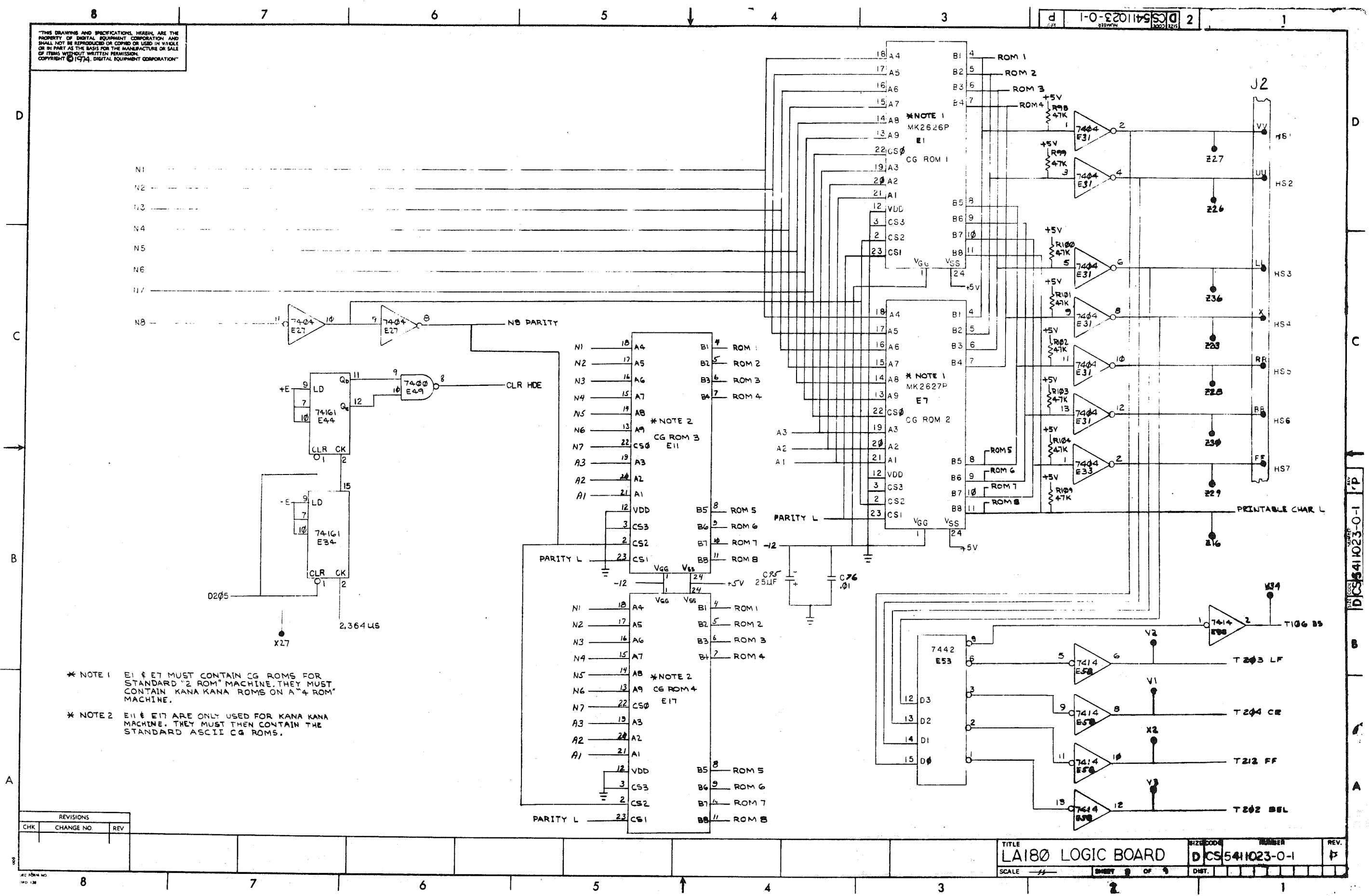
1-0-E201179 500 2



REV.	NO.	DATE
1	1	1970

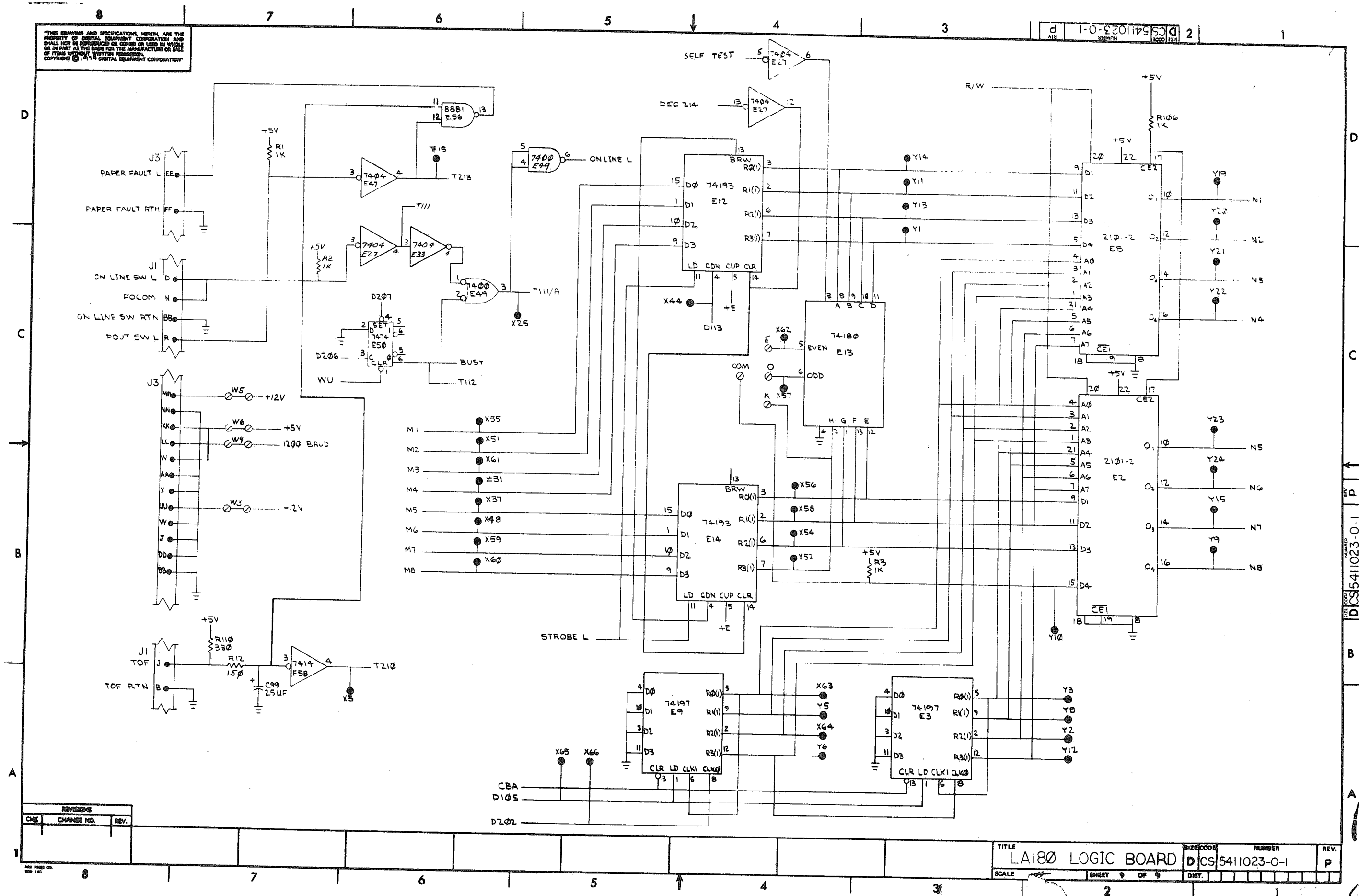
TITLE	NUMBER	REV.
LA180 LOGIC BOARD	DCS 5411023-0-1	P
SCALE	SHEET 7 OF 9	

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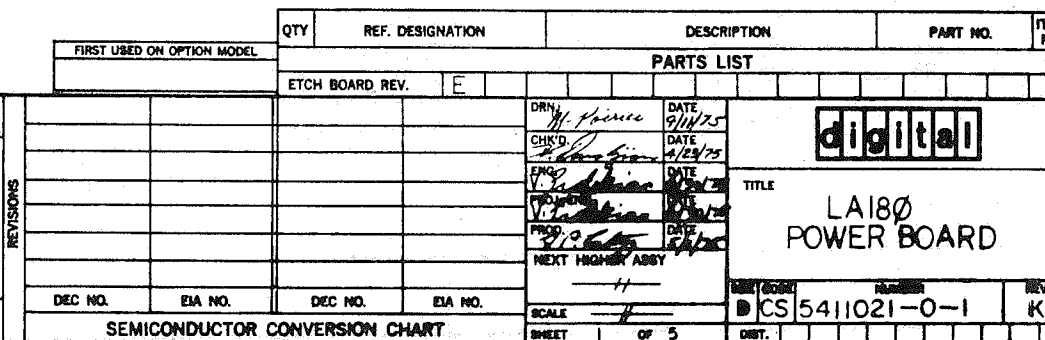


* NOTE 1 E1 & E7 MUST CONTAIN CG ROMS FOR STANDARD "2 ROM" MACHINE. THEY MUST CONTAIN KANA KANA ROMS ON A "4 ROM" MACHINE.

* NOTE 2 E11 & E17 ARE ONLY USED FOR KANA KANA MACHINE. THEY MUST THEN CONTAIN THE STANDARD ASCII CG ROMS.



3. CONNECTORS:
J1 - J1 LOGIC
J2 - FILTER CAPS
J3 - LINE VOLTAGE
J4 - CARRIAGE MOTOR
J5 - LINE FEED MOTOR
J6 - HEAD CABLE



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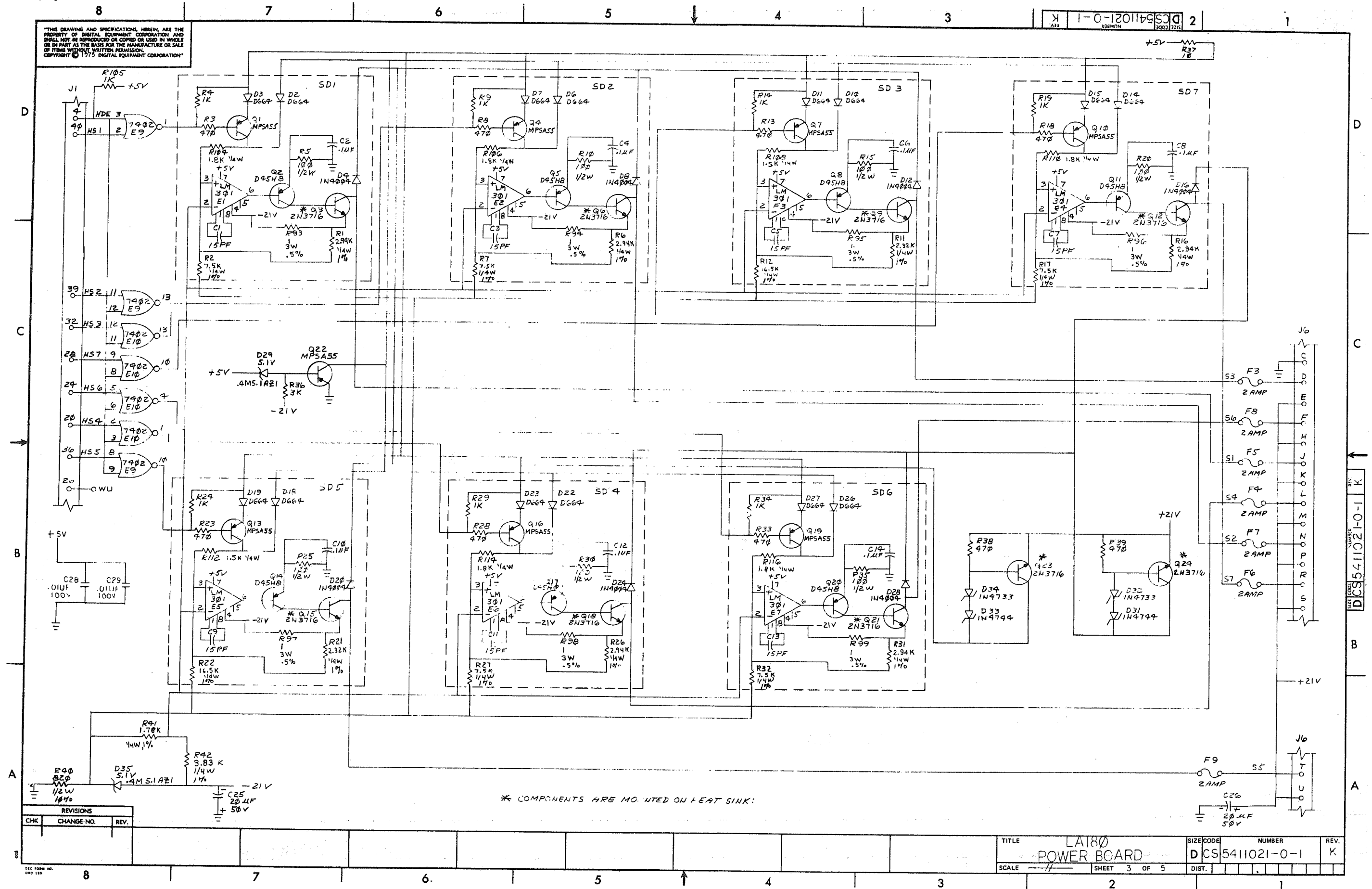
DCS 5411021-0-1 K

PARTS LIST				
QTY	REF DESIGNATIONS	DESCRIPTION	PART NO	ITEM NO
REF		X-Y COORDINATE HOLE LOCATION	KC0-541101-0-4	1
REF		RESISTOR DRILLING HOLE LAYOUT	EAH-541101-0-5	2
REF		MODULE ECO HISTORY	B-MH-5411021-0-6	3
1		ETCHED CIRCUIT BOARD	511020	4
3	C19, C21, C22	CAP 25UF, 25V, 10% +75% HLEL	1000075-00	5
6	C16, C17, C28, C29-C31	CAP .01UF, 15V, 20% DISC	1001610-01	6
1	C19	CAP .22UF, 15V, 10% MIL-C	1000037-00	7
1	C20	CAP 150UF, 12V, 20% AL EL	1000993-00	8
2	C25, C26	CAP .01UF, 5V, 10% +75% HLEL	1002639-00	9
1	C29	CAP .15UF, 35V, 10% MIL-C	1000125-00	10
7	C13, C5, C7, C9, C11, C13	CAP .15UF, 100V, 5% D.M.	1002427-00	11
7	C4, C9, C6, C8, C10, C12, C14	CAP .01UF, 15V, 20% DISC	1000030-00	12
1	C23	CAP .33UF, 25V, 10% J-TANT	1005328-00	13
22	D2, D3, D6, D7, D10, D11, D14, D15, D18, D19, D21, D27, D28, D33, D38, D40, D41, D33, D52, D61, C28, D62	DIODE D662	1100114-00	14
2	D31, D33	DIODE IN4744	1105499-00	15
13	D4, D8, D12, D16, D20, D28, D40, D47, D49, D53, D55, D51, D24	DIODE IN4004	1105796-00	16
4	D54 THRU D57	DIODE IN4721	1110182-00	17
1	D53	DIODE BP105E 21V	1110714-00	18
4	D29, D35, D59, D70	DIODE .4M 5.1A 21.5V, 1% ZENER	1105873-00	19
2	D32, D34	DIODE IN4733	1110488-00	20
1	D20	DIODE IN749A	1104860-00	21
2	D42, D48	DIODE IN749A	1100122-00	22
4		KEYING PLUG, MATE-N-LOK	1211595-00	23
4	J2, J3, J4, J5	CONN MATE-N-LOK 3 SKT	1309340-00	24
28		LOCKET PL. MATE-N-LOK	1209456-01	25
2	R52, R60	RES 2K, 1/4W, 5%	1302388-00	26
1	R47	RES 20, 1/2W, 5%	1302336-00	27
7	R93 THRU R99	RES 1, 3W, 5% W.W.	1312547-00	28
4	R100 THRU R103	RES 4, 5W, 5% W.W.	1311624-00	29
1	R67	RES 100, 1/4W, 5%	1300229-00	30
1	R76	RES 18, 1/4W, 5%	1302124-00	31
2	R73, R74	RES 100, 10W, 5% W.W.	1310597-00	32
7	R5, R10, R15, R20, R25, R30, R35	RES 100, 1/2W, 5%	1300228-00	33
5	R44, R58, R92, R108, R112	RES 1.5K, 1/4W, 5%	1300391-00	34
1	R84	RES 220, 1/4W, 5%	1300271-00	35
2	R56, R64	RES 150, 2W, 5% W.W.	1305524-00	36
2	R55, R63	RES 150, 5W, 5% W.W.	1301898-00	37
1	R83	RES 270, 1/4W, 5%	1301972-00	38
1	R50	RES 390, 2W, 5%	1301864-00	39
1	R75	RES 270, 2W, 10%	1300288-00	40
1	R82	RES 300, 2W, 5%	1309855-00	41
2	R78, R81	RES 390, 1/4W, 5%	1300309-00	42
1	R79	RES 470, 2W, 5%	1303062-00	43
2	R85, R86	RES 150, 1/4W, 5%	1300250-00	44
2	R57, R59	RES 300, 1/2W, 5%	1300291-00	45
9	R43, R45, R9, R4, R19, R24, R29, R34, R105	RES 1K, 1/4W, 5%	1300365-00	46

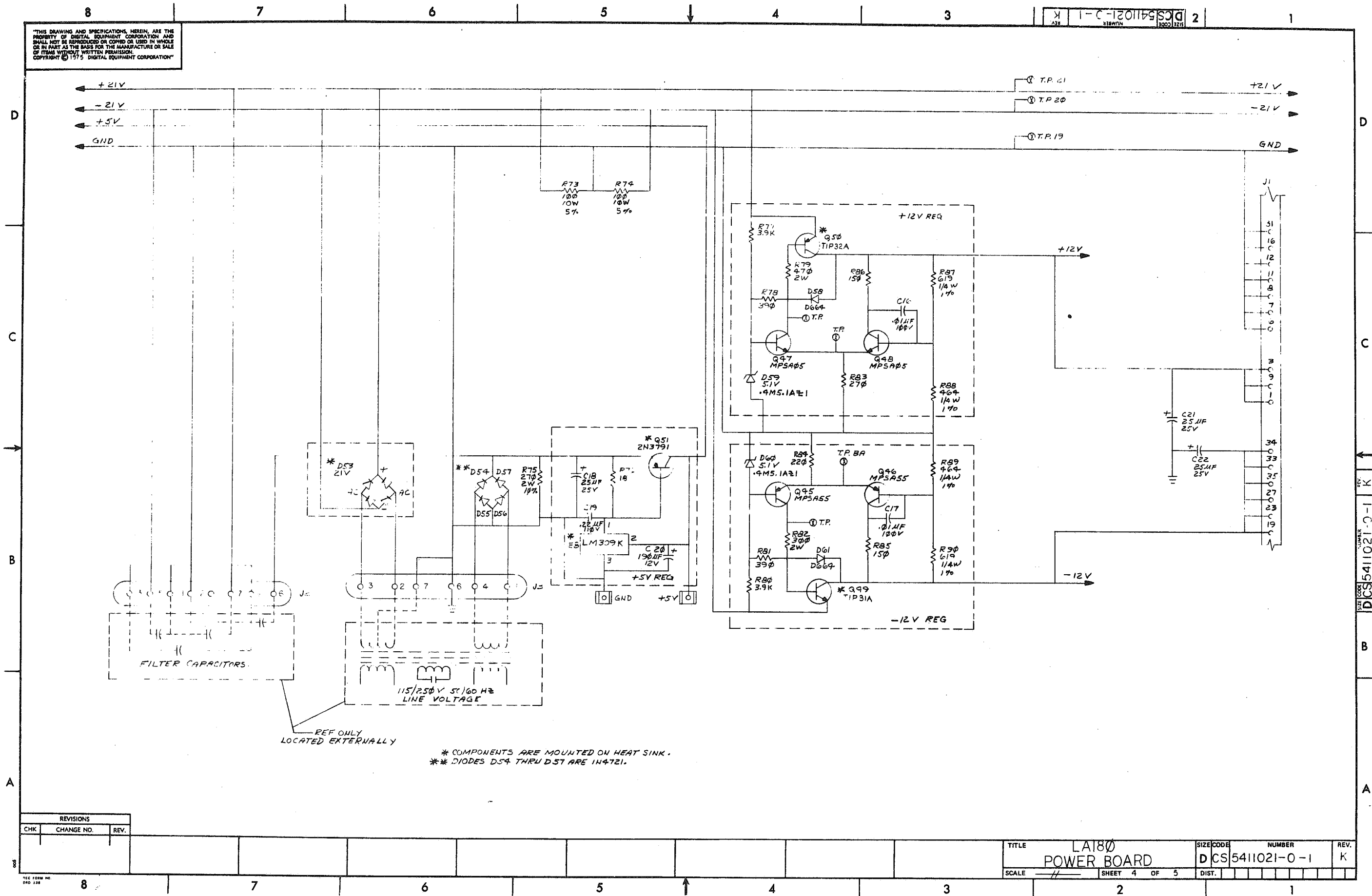
PARTS LIST				
QTY	REF DESIGNATIONS	DESCRIPTION	PART NO	ITEM NO
2	R45, R48	RES 5.1K, 1W, 5%	1303048-00	47
3	R77, R60, R91	RES 3.3K, 1/4W, 5%	1300444-00	48
2	R92, R99	RES 4.7K, 1/4W, 5%	1300397-00	49
3	R53, R61, R66	RES 10K, 1/4W, 5%	1300479-00	50
2	R88, R89	RES 4.0K, 1/4W, 1%	1303047-00	51
2	R97, R90	RES 619, 1/4W, 1%	1305126-00	52
15	R3, R8, R13, R18, R23, R33, R38, R28, R39, R46, R49, R54, R57, R62, R65	RES 470, 1/4W, 5%	1300316-00	53
1	R40	RES 820, 1/2W, 10%	1300353-00	54
1	R30	RES 3K, 1/4W, 5%	1300432-00	55
5	F2, F3, R7, R27, R32	RES 7.5K, 1/4W, 1%	1305322-00	56
1	R41	RES 1.78K, 1/4W, 1%	1302612-00	57
5	R1, R2, R16, R26, R31	RES 2.94K, 1/4W, 10%	1301981-00	58
1	R42	RES 3.83K, 1/4W, 1%	1309413-00	59
1	F1, F4	RES 2.32K, 1/4W, 1%	1312545-00	60
1	G3, G6, G9, G12, G5, G16, G20, G24, G21	TRANS 2N3716	1503069-00	61
1	G42, G43	TRANS 3715	1503068-00	62
1	G51	TRANS 2N3791	1509581-00	63
2	G29, G39	TRANS 6531B	1509338-00	64
6	G66, G30, G39, G47, G48, G33	TRANS MPS 405	1510705-00	65
13	G37, G35, G1, G4, G7, G10, G13, G16, G19, G27, G22, G44, G45	TRANS MPS 455	1510706-00	66
5	G41, G28, G30, G49, G32	TRANS TIP31A	1512590-00	67
4	G40, G29, G31, G50	TRANS TIP32A	1512589-00	68
7	G2, G5, G8, G11, G4, G17, G20	TRANS D45HB	1510708-00	69
2	E9, E10	IC 7402	1909004-00	70
1	E8	IC LM309K REGULATOR	1910460-00	71
7	E1 THRU E7	IC LM301A OP AMP	1910282-00	72
1	J6	HB02 18 PIN (MODIFIED)	5510831-00	73
2		RIVET, POP	3000460-00	74
1		HEAT SINK	7413424-00	75
4		FUSE CLIP	9009513-01	76
9		WASHER (MATE-N-LOK STANDARD)	9006693-00	77
8		EYELET	9006732-00	78
2	F1, F2	FUSE 3 AMP SLO BLO	9007218-00	79
2		FASTON TAB	9007113-00	80
2		EYELET	9009000-00	81
26		SCREW PPH 4-40X1/2	9006013-01	82
35		NUT, KEP 4-40	9006557-00	83
1		SCREW, PPH 6-32 X15/16	9007838-01	84
1		NUT, KEP 6-32	9006560-00	85
1		CABLE, POWER BOARD	6-AD-7010363-1	86
13		INSULATOR, TD-3	9006721-00	87
4		INSULATOR	9009597-00	88
4		COMPOUND, THERMAL JOINT	9008268-00	89
5		SCREW PPH 4-40 X7/16	9006012-01	90
1	F3 THRU F9	FUSE, PICO, 2 AMP	1211751-00	91
1		LABEL	3611567-00	92
5	R104, R106, R110, R114, R116	RES 1.8K, 1/4W, 5%	1300398-00	93
2	R12, R27	RES 16.5K, 1/4W, 1%	1312546-00	94
14	WIRE	WIRE, AWG 14, GREEN	9107370-55	95
2	R10, R71	RES. 60, 10W, 5%	1312997	96

PARTS LIST				
QTY	REF DESIGNATION	DESCRIPTION	PART NO	ITEM NO
1	R37	RES 10 1/4W, 5%	1301317-00	97
2	R117, R118	RES 100 1/4W, 5%	310981-00	98

REVISIONS		
CHK	CHANGE NO	REV

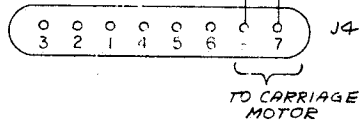
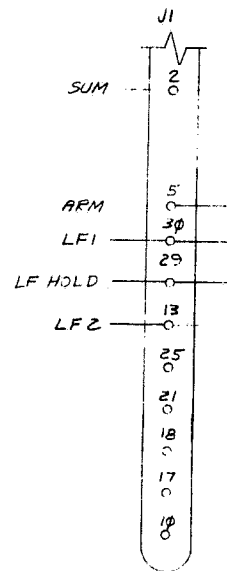


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DCS 5411021-0-1



* COMPONENTS ARE MOUNTED ON HEAT SINK.

REVISIONS		
CHK	CHANGE NO.	REV.

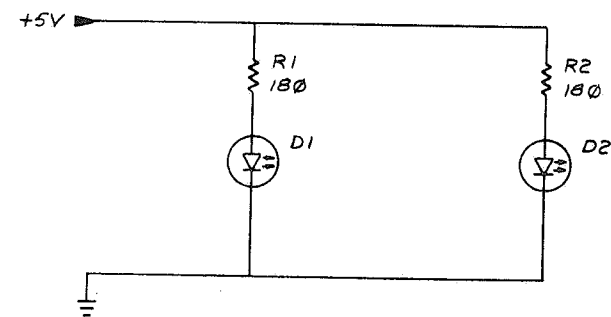
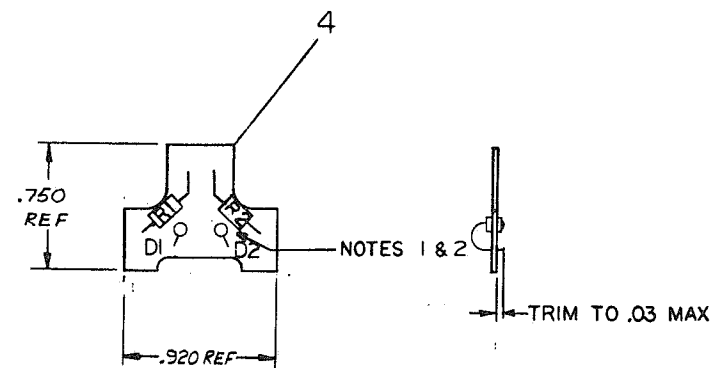
TITLE		SIZE		CODE		NUMBER		REV.	
LA180 POWER BOARD		D		CS		5411021-0-1		K	
SCALE		SHEET		5		OF		5	
		DIST.							

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NOTES:

1. HEAT APPLIED TO D1 AND D2 DURING SOLDERING SHALL NOT EXCEED 285°C FOR A PERIOD OF 15 SECONDS.
2. FINISHED HOLE SIZE FOR MOUNTING D1 AND D2 SHALL BE .078 MIN. .083 MAX

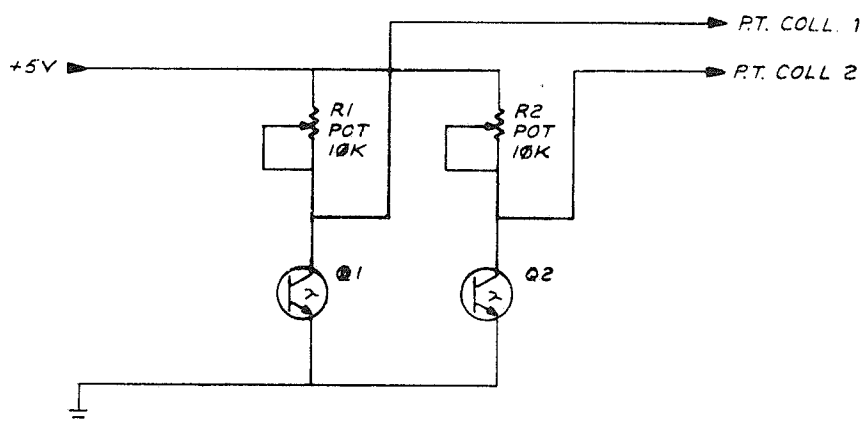
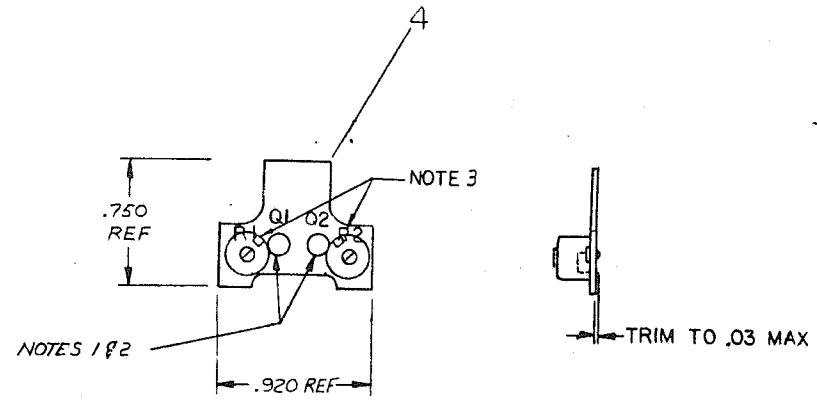


REF		DESCRIPTION	PART NO.	ITEM NO.
REF		X-Y COORDINATE HOLE LOCATION	K-CD-5410991-0-4	1
REF		ASSY/DRILLING HOLE LAYOUT	D-RH-5410991-0-5	2
REF		MODULE ECO HISTORY	B-MH-5410991-0-6	3
1		ETCHED CIRCUIT BOARD	DIA-5010991-0-0	4
2	R1, R2	RES 180 1/4W 5%	1301322	5
2	D1, D2	LIGHT EMITTING DIODE	1111907	6

QTY	REF DESIGNATION	DESCRIPTION	PART NO.	ITEM NO.
PARTS LIST				
ETCH BOARD REV B				
REVISIONS				
DEC NO. EIA NO. DEC NO. EIA NO.				
SEMICONDUCTOR CONVERSION CHART				
TITLE LA36 ENCODER PC BOARD LED				
D/C S 5410991-0-1 G				

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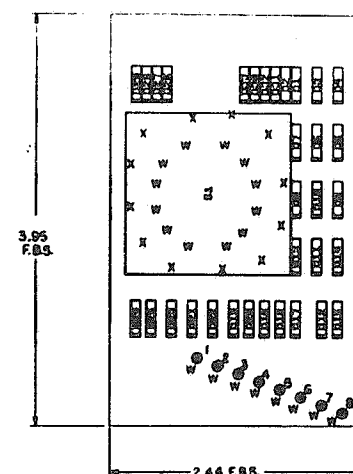
- NOTES:
1. HEAT APPLIED TO Q1 AND Q2 DURING SOLDERING SHALL NOT EXCEED 285° C FOR A PERIOD OF 15 SECONDS.
 2. FINISHED HOLE SIZE FOR MOUNTING Q1 AND Q2 SHALL BE .078 MIN., .083 MAX.
 3. R1 AND R2 MUST BE ORIENTED AS SHOWN, WITH NOTCH TOWARD CENTER OF BOARD.



REF		X-Y COORDINATES & HOLE LOCATION	K-CO-5411049-0-1	1
REF		ASSY/DRILLING HOLE LAYOUT	D-AH-5411049-0-5	2
REF		MODULE ECO HISTORY	B-MH-5411049-0-6	3
REF		ETCHED CIRCUIT BOARD	D-IA-5411049-0-0	4
Z	R1, R2	POTENTIOMETER 10K	1309150-12	5
Z	Q1, Q2	PHOTOTRANSISTOR	1511906	6

QTY	REF DESIGNATION	DESCRIPTION	PART NO.	ITEM NO.
PARTS LIST				
ETCH BOARD REV B				
DRN. <i>[Signature]</i>		DATE 5/2/74		
CHK'D. <i>[Signature]</i>		DATE 5/1/74		
ENG. <i>[Signature]</i>		DATE 2/10/74		
PROD. <i>[Signature]</i>		DATE 2/15/74		
NEXT HIGHER ASSY				
DEC NO.		EIA NO.	DEC NO.	EIA NO.
SEMICONDUCTOR CONVERSION CHART				
SCALE		SHEET 1 OF 1		
TITLE		LA36		
ENCODER PC BOARD		PHOTOTRANSISTOR		
DICS 5411049-0-1		REV. B		

NOTES:



R		SPLIT LOG	9006735-00	7
33	DI-D33	DIODE	1100114-00	7
I	SI	SWITCH SINGLE POLE-12 POSITION	1212443-00	5
I		ETCHED CIRCUIT BOARD	CMD-501622-00	1
REF		MODULE ELECTRONIC	R-MH541613-C-E	3
REF		ALUMINUM PLATE PLATINUM	D-AH-541155-3-0-5	
REF		CRYSTAL UNIT	541155-3-0-4	1
QTY	REF. DESIGNATION	DESCRIPTION	PART NO.	ITEM NO.

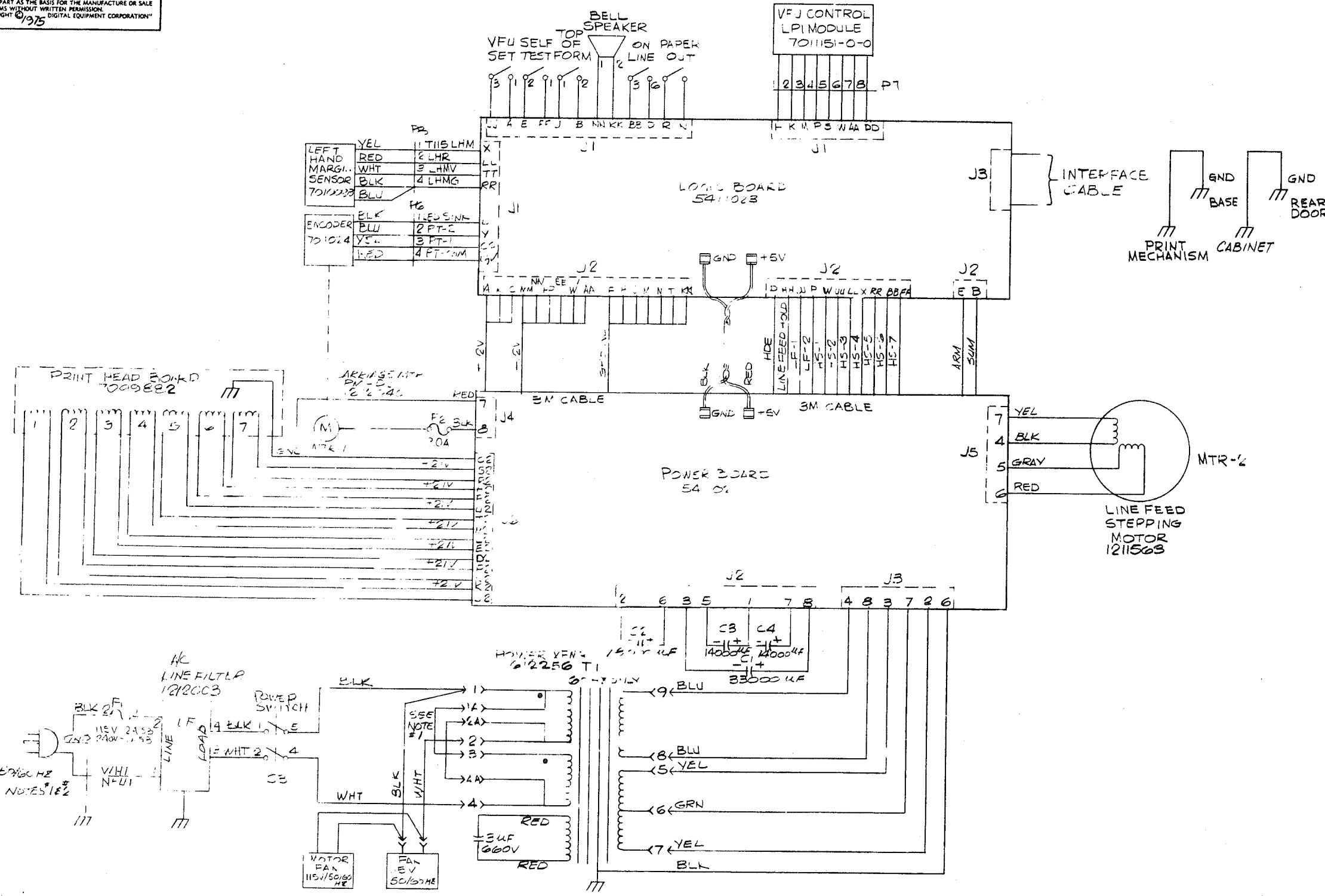
FIRST USED ON OPTION MODEL		QTY		REF. DESIGNATION		DESCRIPTION		PART NO.		NO.	
LA180		ETCH BOARD REV.		A		PARTS LIST					
				DRN.		DATE					
				CHK'D.		DATE					
				APPROVED		DATE					
				PROJ. ENG.		DATE					
				PROD.		DATE					
				NEXT HIGHER ASSY			TITLE		CONTROL		
SEC NO.	EIA NO.	SEC NO.	EIA NO.	SCALE NONE				DWG CODE	NUMBER		REV.
				SHEET 1 OF 1				CS	5411653-0-1		*
SEMICONDUCTOR CONVERSION CHART				DIST.							

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0-0-291110Z
01012
2

NOTES:

1. JUMPERS SHOWN ARE FOR 115V OPERATION. FOR 240V OPERATION REMOVE BOTH JUMPERS ADD ONE JUMPER FROM 2A TO 3
2. FOR 50 HZ OPERATION TRANSFORMER MUST BE REPLACED BY 50 HZ TRANSFORMER #1612257



DRN	<u>1167</u>	<u>4/11/75</u>	FIRST USED ON	<u>LA180</u>	digital A	
CHK'D	<u>1167</u>	<u>4/10/75</u>	TITLE			
ENG.	<u>V. J. Adkins</u>	<u>4/23/75</u>	INTERCONNECTION			
PROJ. ENG.	<u>S. Adkins</u>	<u>4/23/75</u>	DRAWING			
PROD.	<u>1167</u>	<u>4/23/75</u>				
NEXT HIGHER ASSY.						
E-UA-LA180-0			SIZE	CODE	NUMBER	REV.
SCALE <u>1/1</u>			D	IC	7011167-0-0	A
SHEET	<u>1</u>	OF	<u>1</u>	DIST.		

REVISIONS	NO.	REV.
1	1	A

DIGITAL EQUIPMENT CORPORATION
MAYNARD MASSACHUSETTS

PACKAGING INSTRUCTION

DECwriter

REV: A
B

DATE: 7/75
1/76

Quantity	Purchase Specification No.	Description
1	9905601	Stitched Tube
1	9905601-01	Full Telescope Cap
1	9905639	Pad with Foam (LA36/LA35 only)
1	9905129-39	Poly Bag
1	9905651	Wood Skid
1	9905651-01	Wood Brace (2 X 3)
2	90-09288	Tee Nut
2	90-09290-03	3/8-16 X 3 Hex-Head Bolt
1	36-12520-0-0	Caution Tag
6 ft	90-09634	Stain Resistant Tape (3M)
1	90-07032	Panduit Cable Tie
20 ft	9905734-02	Polypropylene Strapping
1	9905129-35	Poly Bag

PACKAGING INSTRUCTIONS

Step	Procedure
1.	Lift the cover and slide the print head to the left as far as it will go. Slip the Panduit Cable Tie (90-07032) around the ribbon rollers (A and B). Do not over-tighten. Fasten Caution Tag (36-12520-0-0) to tractor drive shaft. Close cover.
2.	Place Pad with Foam (9905639) (foam side down) on top of the LA36/LA35 keyboard. Tape with two (2) strips of glass filament tape (90-09634). Tape back door closed, using two (2) strips of tape.
3.	Place DECwriter on top of Skid Deck (9905651).
4.	Place all hardware and software into Poly Bag (9905129-35) and staple to deck of skid.
5.	Place one strip of CW21 (Custom Wrap) or Microfoam, over each leg of the DECwriter.
6.	Place Wood Brace (9905651-01) on top of legs as far back as it will go to properly align the bolt holes.
7.	Place Hex-Head Bolts (90-09290-03) through the bolt holes in the wood brace; place the Tee Nuts (90-09288) up and underneath the skid deck; then tighten down.
8.	Place a Poly Bag (9905129-39) over the DECwriter. Then, place the Stitched Tube (9905601) over and around the unit. Assemble Full Telescope Cap (9905601-01) and place on top of Stitched Tube.
9.	Strap in two places as shown in Figure 2. Use Polypropylene strapping (9905734-02).

NOTE
Make sure that the Stitched Tube is resting on the wood skid base on all 4 sides.

ENG
R. M. McLean
12/24/75
APPD
S. W. P. 12/24/75

SIZE
A

CODE
SP

NUMBER
3700131-0-0

REV
B

DEC 8-1981-1031-1-4671

DRA - 129

SHEET 1 OF 3

PACKAGING INSTRUCTION

DECwriter

REV: _____
DATE: _____

NOTE:
Make changes to "C" size original only and rephotograph.

ENG. _____
APPD. _____
SIZE 1/1
CODE SP
NUMBER 3700131-0-0
REV B

ENC-108
SHEET 2 OF 3

PACKAGING INSTRUCTION

DECwriter

REV: _____
DATE: _____

NOTE:
Make changes to "C" size original only and rephotograph.

ENG. _____
APPD. _____
SIZE 1/1
CODE SP
NUMBER 3700131-0-0
REV B

ENC-108
SHEET 3 OF 3



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CUSTOMER PRINT SET INDEX

THIS IS PRINT SET

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SEQUENCE

SEQUENCE

B-DD-LAXX-KJ
E-UA-LAXX-KJ-Ø
A-SP-LAXX-KJ-2

[illegible][illegible]

